

Evaluating the Geographic Transfer of a US-Trained Surface-Parking Segmentation Model: Evidence from Leeds

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Abstract

The United Kingdom holds no consistent spatial record of off-street surface parking, so the land it occupies is missing from densification debates — even as national planning policy names car parks explicitly among the under-utilised land authorities should bring forward. Segmentation of aerial imagery offers a route needing neither institutional records nor fieldwork, and a model trained for the task has been published, but whether it transfers to British cities has not been tested.

This dissertation applies that US-trained model, exactly as released and with no UK training data in the primary analysis, to 100 km² of Leeds, evaluating it against 2,037 manually labelled car parks drawn to the source model’s target definition. Rather than reporting accuracy alone, it attributes error against independent reference layers, characterises it through a stratified sample of 142 image chips, and tests the building- and road-subtraction stage by ablation.

Transfer proves asymmetric. Recall is 0.854 and less spatially variable than precision, which is 0.571; predicted area is 1.50 times the labelled area. Error is concentrated in boundary placement rather than misrecognition: automated attribution leaves 2.1% of labelled area as unresolved whole-lot FN, while a separate sampled reference-side correction is 1.0%. The building-subtraction stage creates a blind spot of its own, deleting four fifths of the rooftop parking retained in the pre-subtraction output.

Measured against that reliability, the labelled reference puts surface parking at 3.26% of the study area, concentrated in the inner 2 km and declining sharply beyond it. Within Leeds, local calibration corrects over-prediction to about $\pm 7\%$ at half-city scale, though not at 1 km². A bounded fine-tuning supplement is reported separately. The transferred map does not support uncorrected area measurement, but local calibration makes aggregate estimation possible within the city; transfer between cities remains untested.

Declaration

I hereby declare that this dissertation is all my own original work and that all sources have been acknowledged. It is 11,953 words in length.

Signed:

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Acronyms and Abbreviations

CASA	Centre for Advanced Spatial Analysis, UCL
FN	False negative — labelled parking the model did not predict
FP	False positive — predicted parking that is not labelled parking
GSD	Ground sample distance
IoU	Intersection over union
MHCLG	Ministry of Housing, Communities and Local Government
NIR	Near infrared
NPPF	National Planning Policy Framework
OS	Ordnance Survey
OSM	OpenStreetMap
RQ	Research question
TP	True positive — predicted parking that is labelled parking
VOA	Valuation Office Agency

Chapter 1

Introduction

1.1 Context and motivation

Surface parking is widespread in cities but rarely inventoried consistently. It is usually treated as a transport question — how many spaces, priced how, used when — but its more consequential character is spatial. A minimum parking requirement is a rule about how much land a building must set aside, so the cumulative effect of decades of such rules is a pattern of land use, one that generates little activity and no housing, and for which no single agency is typically responsible for inventorying (Shoup, 2005).

In England that omission has become awkward. The National Planning Policy Framework asks planning authorities to bring forward under-utilised land, and names car parks explicitly among its examples (MHCLG, 2024, para. 125(d)). The instruction presupposes an answer that England lacks consistently: where are they, and how much land do they hold? The densification literature runs into the same wall. Britain’s largest cities are found to carry substantial density gaps relative to their European peers, driven in part by the neighbourhoods lying just beyond the city centre (Lange, Kovacevic and Johnson, 2026), yet no consistent measurement exists of how much of that inner land is currently held as car park.

The gap is not for want of trying. It reflects how parking has been measured where it has been measured at all. American inventories are the fullest available, but they are built from cadastral records and codified parking requirements that the UK does not hold in comparable form (Scharnhorst, 2018; Hoehne et al., 2019). British measurement has been local and survey-based — most substantially a sample of on-the-ground inspections in London — and counts *spaces* rather than land area. The authors of the most substantial recent UK review conclude bluntly that little information is collected about how much parking exists at all (Bates and Leibling, 2012).

Segmentation of aerial imagery offers a way past both obstacles, because it depends on no institutional record and no fieldwork. A model trained specifically for parking has been published and can be run on any imagery a user holds (Qiam, Devunuri

and Lehe, 2025). That is an attractive proposition for a planner in a British city, and it is the proposition this dissertation tests.

The risk is that models transfer badly. A network trained on one country’s imagery and urban form may perform quite differently on another’s, a problem sufficiently recognised that benchmarks exist to measure it (Maggiori et al., 2017; Lyu et al., 2025). Nor does the architecture’s own robustness claim settle the matter: SegFormer’s reported zero-shot robustness is demonstrated against corrupted images of familiar cities, which is close to the opposite of the situation here (Xie et al., 2021). Before a transferred map is used as evidence about land, someone has to establish what it can be used for.

That last phrase names an established question rather than a new one. Spatial data quality is conventionally divided into *internal* quality, meaning how far a dataset matches the ground, and *external* quality or fitness for use, meaning whether it is adequate for a particular purpose — a distinction that exists because the same dataset can be sufficient for one task and useless for another, so quality cannot be settled independently of the use (Devillers et al., 2007). This dissertation is a fitness-for-use assessment of a transferred model, and its contribution is to make that assessment quantitative: not whether the map is good, but which uses its measured error still permits.

1.2 Research questions

-
- RQ1** How accurate is a US-trained surface-parking segmentation model on UK aerial imagery, and does that accuracy vary systematically within the city?
- RQ2** What systematic errors does it make, and how much of that error is removed — or created — by post-processing?
- RQ3** Under the measured reliability, how much central-city land is surface parking, and where is it concentrated?
-

The three are sequential rather than parallel. RQ1 establishes whether the transfer works at all. RQ2 explains why the accuracy figures look as they do, which determines what they mean. Only under the reliability those two establish does RQ3 become answerable.

1.3 Scope

The study covers 100 km² of Leeds, validated against 2,037 manually labelled car parks. One city is examined in depth rather than several in outline because comparing cities presupposes that the model’s output is trustworthy, and the literature reviewed here had not established that in the UK; spending the available effort on a labelled reference, a per-cell validation and a sampled error typology establishes what the map supports before anything is built on it. Multi-city extension then becomes future work with a defined precondition, which §5.5 specifies.

Three boundaries define the primary analysis. The target is off-street surface parking as the source model defines it, so on-street parking and enclosed structures are outside it by rule. The model is used exactly as released, with no UK training data, because what is being measured is what an off-the-shelf user would obtain; a bounded supplementary experiment departs from this second boundary alone and is reported separately in Appendix C. And the study identifies where parking is and how much land it occupies; it does not judge whether any site should be redeveloped, a question requiring ownership, access, demand and viability information that none of the data used here contains.

1.4 Ethical considerations

All data are secondary and either openly licensed or licensed to the institution, and no human participants are involved. Two points warrant reflection nonetheless. The labelling was carried out by a single annotator, so the reference embodies one person’s judgement on a target whose boundaries are genuinely ambiguous. Second, an output identifying low-intensity urban land could be read as a development prospectus. The scope statement above is a substantive commitment, not a formality: no site-level claim is made anywhere in what follows.

1.5 Structure of the dissertation

Chapter 2 brings together four literatures — parking as urban land, the evidence on its extent, segmentation of aerial imagery, and the limits of volunteered reference data — and states the gap where they meet. Chapter 3 sets out the annotation protocol, the pipeline, the accuracy measures, the two-track error typology, the ablation design and the checks on imagery consistency and on the calibration estimator. Chapter 4 reports the measured accuracy, decomposes the error, tests the building and road subtractions, shows the apparent location effect to be confounded, and

gives the extent and distribution of surface parking. Chapter 5 answers the three research questions, states what the transferred map does and does not support, and sets out the limitations. Chapter 6 concludes.

Chapter 2

Literature Review

This chapter brings together four literatures that rarely meet. Section 2.1 sets out why the land occupied by surface parking is worth measuring, and why UK policy has recently made that question sharper. Section 2.2 reviews what is actually known about parking extent, and shows that the evidence base and the methods behind it are predominantly American. Sections 2.3 and 2.4 turn to the method that could close that gap — semantic segmentation of aerial imagery, and the specific model used here. Section 2.5 asks what could go wrong when such a model crosses a national boundary, and converts that question into a set of expectations the results chapter can test. Section 2.6 explains why the obvious alternative reference, OpenStreetMap, cannot serve as ground truth. Section 2.7 states the gap.

2.1 Parking as urban land

The case for treating parking as a land-use question rather than a transport one was made most forcefully by Shoup (2005). Minimum parking requirements, he argues, operate as a hidden subsidy: by obliging developers to supply parking regardless of demand, they raise development costs, encourage driving, and commit land to a use that generates little activity and no housing. The argument’s force is spatial. A parking requirement is a rule about how much land a building must set aside, so its cumulative effect is a pattern of land use — and one that is rarely inventoried, because no single agency is typically responsible for counting it.

That framing has become directly relevant to English planning policy. The National Planning Policy Framework instructs authorities to make “as much use as possible of previously-developed or ‘brownfield’ land” (MHCLG, 2024, paras 124–125) and to give substantial weight to brownfield development within settlements. More pointedly for this study, paragraph 125(d) asks authorities to

promote and support the development of under-utilised land and buildings, especially if this would help to meet identified needs for housing where land supply is constrained and available sites could be used more effectively (for example converting space above shops, and building on

or above service yards, car parks, lock-ups and railway infrastructure).

Car parks are thus named in national policy as an example of under-utilised land. The policy instruction presupposes a spatial answer that England does not hold consistently: where are they, and how much land do they hold?

The same gap appears in the densification literature. Centre for Cities research finds that Britain's largest cities — Manchester, Birmingham, Liverpool, Leeds and Glasgow — carry the largest density gaps relative to their European peers, and that the gap is driven substantially by post-war neighbourhoods “located just beyond the city centre”, which can be up to 40 per cent less dense than similarly located pre-war neighbourhoods (Lange, Kovacevic and Johnson, 2026). Livingstone, Fiorentino and Short (2021) examine how densification policy plays out in practice in London and at what cost, while Habermehl and McFarlane (2025) argue that density is better understood as a contested dialectic between “hard” and “gentle” forms than as a single quantity to be maximised. What none of this work can draw on is a consistent measurement of how much land inside British cities is currently held as surface parking. The policy asks for under-utilised land to be found; the evidence base cannot yet say how much of it is car park.

2.2 What is known about parking extent, and where that knowledge comes from

Systematic parking inventories exist, but they are overwhelmingly American, and they were built by methods that do not port straightforwardly to the UK. The British evidence that does exist measures something different, in one atypical city, by means too expensive to repeat.

Scharnhorst (2018) compiled comprehensive inventories for five US cities — New York, Philadelphia, Seattle, Des Moines and Jackson, Wyoming — combining satellite imagery with tax and cadastral records. The results are striking both for supply and for its use: reviewing occupancy studies across the five cities, he reports empty stalls making up 68 per cent of supply in Jackson's residential core. Hoehne et al. (2019) take a different route for metropolitan Phoenix, cross-referencing cadastral and roadway data against minimum parking requirements to estimate 12.2 million parking spaces in 2017, against 2.86 million registered vehicles.

The British evidence is thinner, and its own authors say so. The most substantial recent review of UK parking policy, Bates and Leibling's (2012) study for the RAC Foundation, concludes that a central obstacle to coherent policy is simply the absence of data: their study, they write, shows “how little information is collected about

the quantity of parking space” that exists (p. 99), a problem they attribute to fragmented responsibility and to local authorities lacking the resources to audit their own parking supply. Where UK measurement has been attempted it has been local and survey-based. The fullest example remains a study commissioned for London, in which parking availability was estimated by inspecting a sample of three hundred 500 m squares on the ground, later partially resurveyed; it put the capital’s supply at roughly 6.8 million spaces, counted as spaces (Bates and Leibling, 2012).

Three limitations of that evidence base define the opening this study works in. It counts spaces rather than land area, which cannot answer a question about how much ground a city gives over to parking. It is concentrated on London, which is atypical of British cities in density, land value and parking regulation alike. And it rests on ground survey, which is expensive enough that the exercise has not been repeated at scale or extended to other cities. No comparable measurement of off-street surface parking *area* appears to exist for a British city outside London.

Two further implications follow for the present study. The first is substantive: available inventories and occupancy studies have generally revealed substantial unused or excess parking capacity, making its land take a live policy question rather than an accounting curiosity. The second, and more important here, is methodological. Existing inventory methods are difficult to reproduce nationally in the UK. Hoehne et al. (2019) combine parcel-level cadastral and roadway data with jurisdiction-specific minimum parking requirements. Scharnhorst (2018) combines tax-assessor and institutional records with city-specific aerial or satellite land-cover data, noting that inconsistent data structures make each city a custom exercise. British evidence remains local and survey-based; Bates and Leibling (2012) report that adequate audits are resource-intensive and that filling the gaps nationally would be difficult and costly. Detecting surface parking directly from imagery therefore removes the need for a national parcel-level parking register or a citywide field survey, although its accuracy must still be assessed against locally annotated reference data.

A related strand of work provides the spatial framing rather than the counts. Jiao (2015) shows that urban land density follows regular, describable functions of distance from the centre, which supplies a natural way to organise a within-city analysis: not “how much parking is there” alone, but how its share of land changes across the urban gradient.

2.3 Segmenting parking from aerial imagery

Semantic segmentation of remote-sensing imagery is now dominated by deep encoder–decoder architectures (Lv et al., 2023). The model used here is built on SegFormer (Xie et al., 2021), whose relevant property for aerial work is that it dispenses with positional encoding, leaving it comparatively indifferent to the tile size it is given — useful because aerial imagery is processed as arbitrary tiles rather than at a canonical size.

Its authors also report strong zero-shot robustness, demonstrated on Cityscapes-C, a benchmark that degrades image quality with noise, blur, weather and digital artefacts while leaving the scene unchanged. That is a different test from the one run here. A corrupted image of a familiar city and a clean image of an unfamiliar one stress different capacities, so the reported robustness does not by itself settle whether the model transfers geographically. Chapter 5 returns to the point once the error has been decomposed and it can be said which capacity the crossing taxed.

Parking is an attractive segmentation target because it is visually distinctive: a paved surface, usually with painted bays, usually with vehicles on it. It is also a target whose boundary is genuinely ambiguous — where a car park ends and its access road, service yard or forecourt begins is a matter of definition rather than observation. That ambiguity is not incidental here. It accounts for a substantial share of the apparent error measured in Chapter 4, and separating it from genuine misrecognition is one of this study’s main analytical tasks.

Two studies take parking specifically as the segmentation target. Berry et al. (2019) address a direct consequence of that ambiguity — adjacent car parks merge into one another under ordinary semantic segmentation — by segmenting instances through associative embeddings rather than classifying pixels. Hurst-Tarrab et al. (2020) assemble APKLOT, a purpose-built dataset of parking-block polygons cut from satellite imagery, and report the accuracy their models reach on it — the closest published point of comparison for a parking segmentation result, obtained with training and test data drawn from the same source, and taken up in §5.1. Neither study asks what the resulting map is fit for; accuracy is the endpoint.

2.4 The model used here and its annotation definition

The model applied in this study is the parking-lot segmentation network released by Qiam, Devunuri and Lehe (2025), who introduce both a pipeline and an NIR-enhanced training dataset for the task. No published accuracy figure is carried over

to describe it. It is used here exactly as released, with no UK training data in the primary analysis.

What matters as much as the architecture is the *definition* the model was trained on, because that definition determines what a correct output looks like. Qiam, Devunuri and Lehe’s target is off-street surface parking visible from above: marked bays and the aisles that connect them, including rooftop parking where the surface is visible — their dataset admits a parking structure only where the deck on top of it is visible — and excluding on-street parking and enclosed structures. Access driveways were not excluded by rule but left to case-by-case judgement, with annotators encouraged to take in only very short ones, expressly so that the model would not learn to recognise roads in general. Boundaries were drawn to the edge of the paving rather than to the parcel line, on the reasoning that only the paved edge offers a visual cue the model can learn; the authors note that OpenStreetMap parking polygons instead follow the parcel. They also note that NIR information assists in separating parking surfaces from adjacent vegetation.

Two consequences run through this dissertation. First, any reference dataset used to assess the model must follow the same definition, or the resulting accuracy figures measure disagreement about categories rather than model performance; the annotation protocol in Chapter 3 is therefore derived from theirs. Second, the imagery available for this study is three-band RGB with no NIR channel — a limitation returned to below.

2.5 Domain shift: what could go wrong across a national boundary

A model trained in one geographical setting frequently performs worse in another — domain shift, surveyed by Lyu et al. (2025) — with sources conventionally divided into sensor and resolution differences, illumination differences, and differences in the objects themselves. The problem is recognised enough that benchmarks exist to measure it. Maggiori et al. (2017) built the Inria Aerial Image Labeling benchmark so that testing is carried out over entirely different cities rather than over held-out parts of the training area, and judged the accuracy their network reached on those unseen cities to represent satisfactory generalisation. Hong et al. (2023) approach the same problem from the other side, arguing that models succeeding within a single city meet a bottleneck across cities and regions.

Two things about that work bear directly on this study. Maggiori et al. operate at 0.3 m with RGB bands and two classes, against 0.25 m RGB and two classes here, so their accuracy levels are a reasonable frame of reference for Chapter 4 rather than

an arbitrary comparison. But those benchmarks test transfer between cities within a broadly shared building stock, whereas the transfer examined here also crosses a national boundary, an imagery programme and a different tradition of laying out car parks.

Most of that literature is concerned with *correcting* domain shift through adaptation. That option is not open to the user this study has in mind. Adaptation presupposes either labelled data in the target domain or substantial engineering to exploit unlabelled data, and a planner who wants to know how much land in their city is car park has neither; their realistic option is to run the published checkpoint on the imagery they hold. The uncorrected output is therefore the object of study here, and the question is what it can still be trusted to do.

Stating that a model was trained in the US and applied in the UK is not on its own an analysis. Table 2.1 names specific differences or working priors that plausibly matter for this target, and states the failure expected if each is relevant.

Difference between the training and application settings	Expected failure
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Table 2.1. Hypothesised sources of transfer error, and the observable failure each would produce if relevant.

Difference between the training and application settings	Expected failure
British off-street car parks are typically smaller and more irregularly shaped than American ones — a prior of this study, drawn from the labelling rather than from a source, and tested rather than assumed	Failures concentrated in small lots and awkward site geometry
A working prior that unmarked parking is more common; the annotation rules accept vehicles plus layout as sufficient evidence	Missed lots lacking painted bay markings
Setts, block paving and gravel are common surfacing materials	Missed lots whose surface is not asphalt
Leeds lies at 53.8 °N, well north of the US cities in the training data (mostly 30–42 °N), so solar elevation is lower and shadows longer; street tree canopy is also denser	Missed or fragmented lots under shadow and canopy occlusion
Commercial vehicle and van share differs	Failures on lots occupied by vans and lorries rather than cars
The imagery used here is RGB only, with no NIR band	Vegetated ground and grass margins confused with paved parking

Each expectation implies a category in the error typology of Chapter 3 and is examined against sampled evidence in Chapter 4. The results can show which mechanisms occur and how frequently, but cannot by themselves establish that a US–UK difference caused them because no comparable source-domain decomposition is available. Setting the expectations out in advance still allows them to be revised: one is substantially revised by the results.

2.6 Why OpenStreetMap cannot serve as ground truth

The obvious way to avoid manual labelling would be to validate the model against OpenStreetMap’s `amenity=parking` features. OSM is the best-known instance of

what Goodchild (2007) named volunteered geographic information: geographic data contributed by non-specialists outside any formal quality-control regime. The quality literature that framing prompted makes clear why the shortcut does not work.

Haklay’s (2010) comparison of OSM against Ordnance Survey data — the foundational UK study — found positional accuracy to be reasonable where coverage exists, but completeness to vary sharply between places, being far better in areas with more contributors. Later reviews confirm that completeness varies both between places and between feature types, though not always in the same direction: Sehra, Singh and Rai (2013) note that the urban–rural gap found in Germany and England ran the opposite way in parts of the United States, where rural coverage was in places more complete than the proprietary datasets it was compared against. Zhou, Wang and Liu (2022) look specifically at land-cover and land-use tagging and find accuracy and completeness varying substantially by category, which matters directly here because parking is recorded as a land-use-like polygon rather than as part of the road network that attracts the most contributor attention.

There is also a definitional problem independent of completeness. As Qiam, Devunuri and Lehe observe, OSM parking polygons are frequently drawn to parcel boundaries rather than to the edge of the paved surface, so even a perfectly complete OSM layer would disagree with an imagery-derived one at the margins.

OSM is therefore treated in this study in two distinct roles, kept strictly separate. Its building and road layers are *inputs* to the post-processing stage of the pipeline. Its parking, land-use and related layers are used only to *attribute* errors after the fact, never to decide what the map should contain. Independently, the manual reference built here allows OSM’s parking completeness to be assessed rather than assumed — a secondary result reported in Chapter 4.

2.7 Research gap

The four literatures leave a gap where they meet. Parking is argued to be under-used urban land, but the empirical evidence for how much land it occupies is overwhelmingly American and rests on institutional records the UK does not hold in comparable form; the British evidence counts spaces rather than area, covers London rather than the country, and its own authors identify the absence of data as the central problem. English planning policy now names car parks explicitly as under-utilised land to be brought forward, and the densification literature identifies exactly the kind of city — and the kind of inner ring — where that land would matter most, but neither can

point to a measurement. Segmentation models can produce such a measurement at scale, and one trained specifically for parking is publicly available, but the parking segmentation literature reports accuracy as an endpoint, and the transfer literature is largely concerned with correcting domain shift rather than characterising the residual reliability of an uncorrected transfer. And the reference dataset that would let a UK user check the output for themselves is incomplete in ways that vary from place to place.

The gap this dissertation addresses is therefore not simply that nobody has mapped parking in Leeds. It is that a scalable method exists, but the literature reviewed here does not establish whether its unadapted UK output is reliable enough to locate parking and estimate its extent. Reporting precision and recall alone does not answer that question. Establishing which errors are boundary effects, which are systematic confusions, which are disagreements about definition, and which are artefacts of the processing pipeline — and then testing whether the resulting aggregate bias is stable enough to be corrected, and at what spatial grain — does. That is the contribution attempted here.

Chapter 3

Methodology

This chapter sets out how the transfer was tested. It describes the study area and the imagery the model consumed (§3.1), the annotation protocol used to build an independent reference (§3.2), the segmentation pipeline itself (§3.3), and the accuracy measures (§3.4). It then sets out the two complementary procedures used to characterise error — automated attribution against reference layers (§3.5) and stratified sampling with visual adjudication (§3.6) — followed by the ablation design used to isolate the contribution of building and road subtraction (§3.7). Two checks close the chapter: one on the imagery underlying the reference and the predictions (§3.8), and one on the estimator used to correct the model’s systematic bias (§3.9).

3.1 Study area and data

The study area is a 100 km² square centred on Leeds, divided into one hundred 1 km² cells on the British National Grid (Figure 3.1). The city centre is taken as City Square (E 429832, N 433449); cell centroids lie between 0.34 km and 7.64 km from it. Leeds was chosen as a large English city outside London with substantial surface parking in and around its core, and with complete aerial coverage at the resolution the model requires. The choice also aligns with the project’s Centre for Cities framing: Leeds is among the large British cities the organisation identifies as having some of the largest density gaps relative to their international peers (Lange, Kovacevic and Johnson, 2026). A square grid rather than an administrative boundary was used so that every cell is an equal-area unit and per-cell statistics are directly comparable. The choice of areal unit is not neutral — figures computed over zones depend on how the zones are drawn, the modifiable areal unit problem (Openshaw, 1984) — so the unit is held constant throughout and whole-area figures are reported alongside per-cell ones.

The imagery is Getmapping aerial photography supplied through Digimap: 100 tiles at 0.25 m ground sample distance, three visible bands (RGB), projected in EPSG:27700. All spatial operations are carried out in EPSG:27700, whose units are metres, so polygon areas are read directly in m² without further projection.

Study area: 100 km² on the British National Grid, Leeds

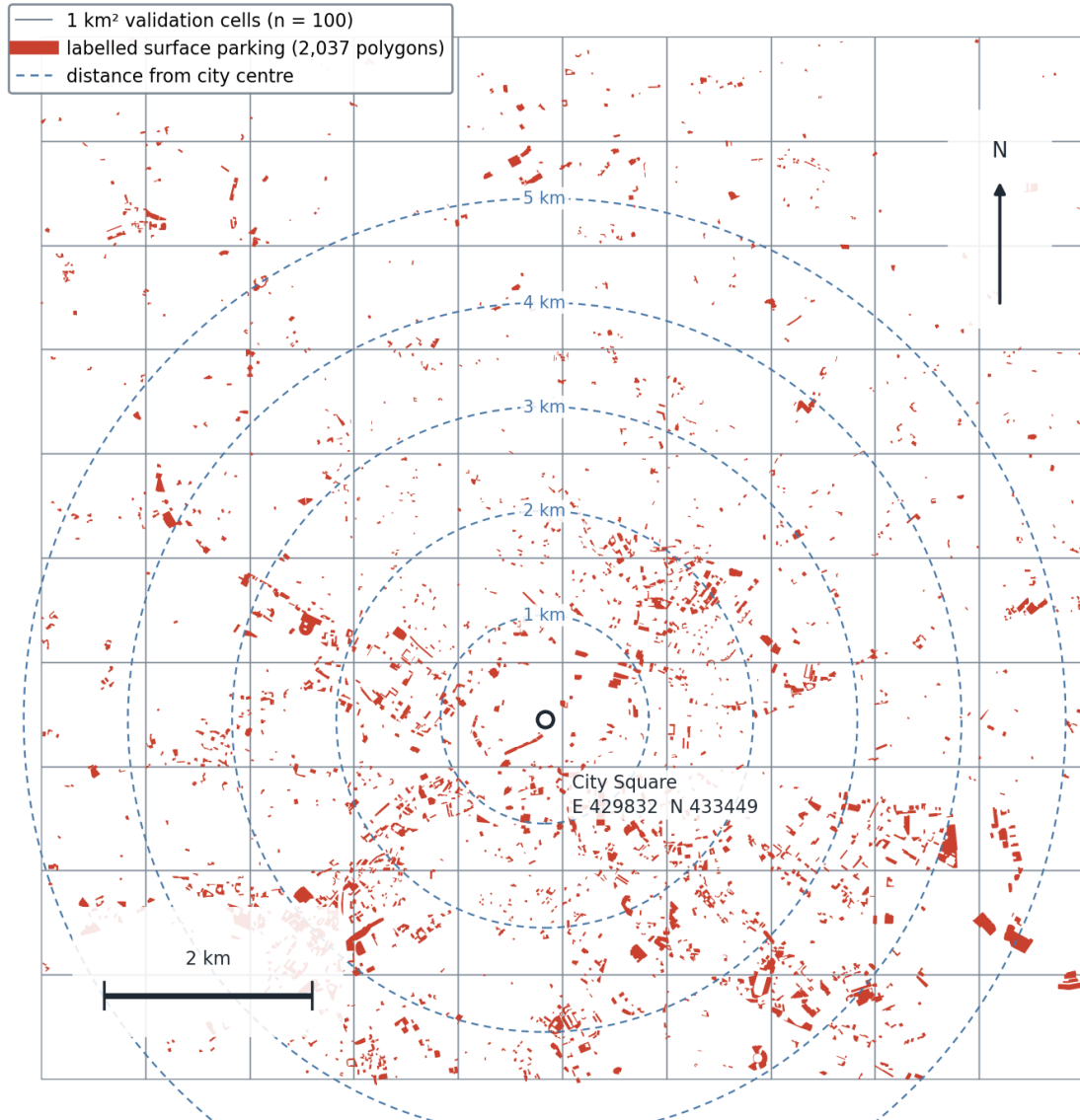


Figure 3.1. The study area: one hundred 1 km² validation cells on the British National Grid, the 2,037 manually labelled surface parking polygons, and distance rings from City Square. Labelled parking is visibly concentrated in a band around, rather than at, the centre — a pattern quantified in §4.7.

Three reference datasets are used, none of them as ground truth. OpenStreetMap building footprints and road centrelines (retrieved 25 June 2026) are inputs to the post-processing stage. OpenStreetMap land use, brownfield, pitch and amenity=parking polygons are used only to attribute errors after the fact. Ordnance Survey Open Greenspace supplies sports facilities. The distinction matters: reference layers are used here to explain where errors fall, and — with the deliberate exception tested in §3.7 — never to decide what the map should contain.

Appendix D lists the code repository, the licensing position of each dataset and the file behind every table reported here. The manual reference labels are available in the repository, but the institutionally licensed aerial imagery cannot be redistributed.

3.2 Annotation protocol

Accuracy figures are only meaningful against a reference that follows the definition the model was trained on. The annotation rules therefore follow those of Qiam, Devunuri and Lehe (2025), whose dataset the model was trained on, and are reproduced in full in Appendix A. The target is off-street surface parking: open-air parking surfaces visible from above, outside the public road. Labels are binary. No minimum size threshold is applied. Marked bays and the aisles connecting them are included, as is rooftop parking where the parking surface is visible from above; on-street parking and enclosed multi-storey structures are excluded. Where markings are absent, an area is labelled only where parked vehicles and a bay-and-aisle layout together make the use unambiguous. Boundaries are drawn at the edge of the paving rather than the parcel line.

One boundary inside residential parking is worth stating explicitly, because it is the commonest ambiguous case in a British city. Parking courts shared between several dwellings are labelled; driveways and forecourts serving a single household are not. A communal court resembles a small car park in layout and in what it does, while a single driveway does not. The distinction is not introduced here: UK parking measurement already separates private residential parking into driveway and communal categories, and the two were surveyed and reported separately in the London study described in §2.2 (Bates and Leibling, 2012).

Labelling was carried out in QGIS by a single annotator over a satellite basemap, producing 2,037 polygons with a confidence attribute (3 = clear, 2 = fairly clear, 1 = uncertain) and a free-text note used to flag rooftop cases. Summed individually the polygons cover 3.2677 km². Dissolving them leaves both the area and the part count unchanged, so no two labelled polygons overlap; the reference is nonetheless

dissolved before every comparison, since the accuracy measures of §3.4 are only well defined on non-overlapping geometry and the property should be enforced rather than assumed. Clipping to the study grid removes 0.0081 km² where polygons cross the outer boundary, giving the 3.2597 km² used throughout. One cell (c0r9) contains no labelled parking, so recall is undefined there and per-cell statistics involving recall use $n = 99$.

Single-annotator labelling remains a limitation, taken up in §5.5.

3.3 Model and pipeline

The model is the SegFormer (Xie et al., 2021) parking-lot segmentation network released by Qiam, Devunuri and Lehe (2025). The released checkpoint is a B5 configuration whose backbone was initialised from Cityscapes weights and fine-tuned by those authors on their parking dataset, as documented in the released model card and repository. In the primary analysis, its weights are used exactly as released: no UK imagery is used to update them. The network is therefore evaluated under zero-shot geographic transfer, although the surrounding pipeline includes the UK-specific tiling and post-processing described below and isolated by the ablation in §3.7. Appendix C reports a separate experiment on a fixed 40/10/50 cell split, comparing generic fine-tuning, positional targeted loss weighting and validation-selected thresholding on raw-pixel outputs without post-processing. Those results remain separate from the primary analysis in §§4.1–4.7.

The pipeline proceeds in five stages. Digimap tiles and their world files are converted to georeferenced GeoTIFFs. Each tile is cut into 512×512 pixel patches, zero-padded at the edges. Patches are passed through the network and the upsampled logits reduced to a binary mask by argmax. Masks are cleaned with a mode filter and vectorised by contour extraction, with components below 1,000 px² (about 62 m² at 0.25 m) discarded and enclosed holes subtracted; polygons are then transformed from pixel to grid coordinates and merged across tiles.

Two post-processing subtractions follow, both UK adaptations of operations already present in the released pipeline, which likewise subtracts building footprints and buffered OpenStreetMap roads (Qiam, Devunuri and Lehe, 2025). Here, OpenStreetMap building footprints replace the source pipeline’s Microsoft US building data and are subtracted under its premise that roofs are false positives — a premise tested against rooftop parking in §3.7. Road centrelines are buffered by carriageway class (Table 3.1), rather than by recorded lane count alone, dissolved, and subtracted.

The half-widths are approximations by class rather than measured carriageway di-

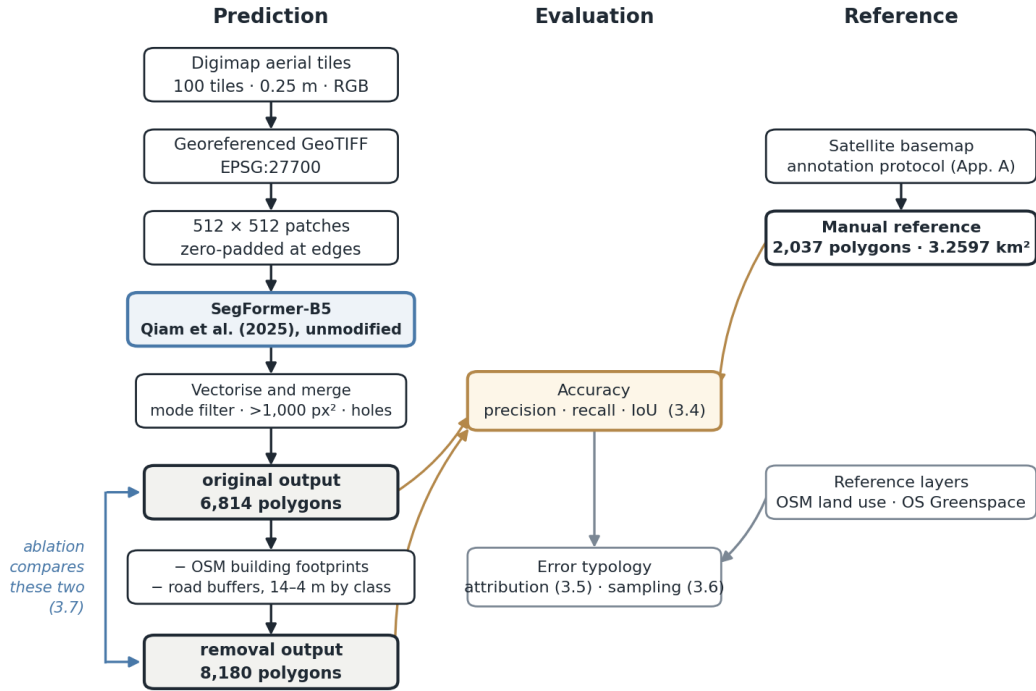


Figure 3.2. The processing chain. Prediction runs from Digimap tile to outputs before and after building and road subtraction, which the ablation of §3.7 compares. The manual reference and the model outputs meet at the accuracy measures of §3.4; the reference layers enter only at the error-attribution stage of §§3.5–3.6.

Table 3.1. Road buffer half-widths by OSM highway class. Where a `lanes` tag is present, the width is the greater of the tabulated value and 3 m per lane.

Class	Width (m)	Class	Width (m)
motorway	14	tertiary	7
trunk	12	tertiary_link	5
primary	10	unclassified	5
secondary	8	residential	5
motorway_link	9	living_street	4
trunk_link	8	default	5
primary_link	7		
secondary_link	6		

mensions: each value is a plausible half-width for the road type the OSM class denotes, with 5 m applied to any class not listed. Footways, cycleways, bridleways, tracks, steps, pedestrian ways and service roads, among others, are excluded from the road layer, service roads in particular because they commonly run *through* car parks; buffering them would remove the aisles the protocol explicitly includes. Both subtractions are evaluated rather than assumed in §3.7.

The output before subtraction (6,814 polygons) and after (8,180 polygons) are both retained. The increase in count reflects single lots being split by the subtracted geometry, which is why polygon counts are never interpreted as numbers of car parks.

3.4 Validation design

Accuracy is measured by area rather than by object, following the general practice in land-cover accuracy assessment of comparing a map against an independent reference over a defined spatial support (Foody, 2002; Olofsson et al., 2014). Let M and R denote the dissolved model and reference geometry, and $|\cdot|$ denote area in m^2 . Dissolving before comparison ensures overlapping geometry is not double-counted. The three quantities are then set operations:

$$\text{TP} = M \cap R, \quad \text{FP} = M \setminus R, \quad \text{FN} = R \setminus M$$

from which

$$\text{precision} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FP}|}, \quad \text{recall} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FN}|}, \quad \text{IoU} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FP}| + |\text{FN}|}$$

Area-based measures were chosen because the study asks how much land the map assigns to parking. Object-based matching would require an arbitrary rule for deciding whether split or merged polygons represent the same lot, particularly after post-processing. Since accuracy estimates depend on the assessment unit, the unit should reflect the intended use of the map (Stehman and Wickham, 2011; Stehman and Foody, 2019). As an internal check, $|M| = |\text{TP}| + |\text{FP}|$ and $|R| = |\text{TP}| + |\text{FN}|$; both hold to four decimal places.

Two aggregations over the n cells are reported. Writing TP_c for the true positive area within cell c , the micro (area-weighted) and macro (cell-averaged) forms of precision are

$$p_{\text{micro}} = \frac{\sum_c |\text{TP}_c|}{\sum_c (|\text{TP}_c| + |\text{FP}_c|)}, \quad p_{\text{macro}} = \frac{1}{n} \sum_c \frac{|\text{TP}_c|}{|\text{TP}_c| + |\text{FP}_c|}$$

and recall and IoU follow the same pattern. The terms are borrowed from classification evaluation, where the same two averages are taken over classes; here they are taken over spatial units. The micro figure treats the whole study area as one unit, so large car parks carry proportionate weight; it answers whether the *total area* is right, and because summing over cells returns the whole-area quantity, it does not depend on how the cells were drawn. The macro figure weights every cell equally regardless of how much parking it contains, and does depend on it. The two are reported side by side because a global figure summarises the map as a whole while masking how accuracy varies across it, and local accuracy is better reported as an accompaniment to the global figure than in place of it (Foody, 2005; Stehman and Foody, 2019). Doing so needs a reference dense enough to support a figure in every sub-region, ordinarily the obstacle to it; here the reference is wall-to-wall rather than sampled, so every cell carries its own. Macro is used as a diagnostic of uniformity rather than as an estimator of any population quantity, and the difference between the two is itself the result: where macro falls below micro, the model is performing worse in cells with little parking than the area-weighted figure suggests.

3.5 Error typology I: automated attribution

The first characterisation of error asks where FP and FN area falls relative to independent layers. It is applied exhaustively to all of it.

Boundary effects are separated first. Area-based measures cannot distinguish a slightly oversized car park from a false detection elsewhere when the error area is the same (Csurka, Larlus and Perronnin, 2013). Boundary IoU restores sensitivity to such errors by evaluating within a fixed contour band, particularly for large objects whose interiors dominate Mask IoU (Cheng et al., 2021). This study adapts that principle to partition, rather than rescore, boundary and non-boundary error.

FP area within a fixed distance d of a labelled lot is boundary *dilation* — the model drawing the same car park slightly too large — as distinct from a standalone false detection elsewhere. Symmetrically, FN area within d of a predicted area is *erosion*: reference area the model did not cover, but lying immediately alongside something it did find.

$$\text{FP}_{\text{dilation}}(d) = \text{FP} \cap (R \oplus d), \quad \text{FN}_{\text{erosion}}(d) = \text{FN} \cap (M \oplus d)$$

where $\oplus$ denotes dilation by a buffer of d metres. Erosion is defined against the *prediction* rather than against the reference boundary, so that it captures only reference area adjacent to a detection; defining it against the reference boundary would also count the outer edge of lots the model missed entirely, which is a detection failure rather than a boundary effect. Because no threshold separates the two cleanly, results are reported at $d = 2, 5$ and 10 m, with 5 m used as the working value. Sensitivity to the width chosen is a known property of band-based measures rather than a peculiarity of this application (Cheng et al., 2021), which is why three are given. That 5 m is a convention rather than a natural break is also visible in the sampled evidence: the three sampled FP fragments that no category explained sit at a median distance of exactly 5.0 m from the nearest labelled lot, against 30.4 – 168.6 m across the substantive categories.

Standalone FP is then attributed in two ways, reported side by side.

- An exclusive partition: reference layers $L_1, \dots, L_k$ are peeled off in sequence, each claiming only the area not already claimed, so shares sum to 100%. Layers are ordered by how precisely they locate the phenomenon — building footprints and sports pitches, which are exact polygons, before road proximity buffers, before broad land-use classes — with industrial and commercial land last because it is the least specific evidence.
- An unordered overlap: each layer is intersected with all FP independently, $FP \cap L_j$, so categories may overlap one another and need not sum to 100%.

Presenting both means the substantive conclusions do not depend on the chosen ordering; as a robustness check, moving industrial land from an early position to last changes its exclusive share from 30.0% to 29.6% . The two have different denominators and are not differences of one another — a point carried into the results tables, where the exclusive column includes the dilation band as its first row so that it sums to 100% .

FN is defined at the level of the car park, not the fragment. An earlier version defined missed parking as reference area more than 5 m from any prediction, which conflated genuinely undetected car parks with the ragged edges of ones the model had largely found. Two diagnostics showed this: of 297 such fragments, 221 lay at a distance of exactly 5.0 m — the threshold itself — and 19.2% of the area belonged to lots the model had detected to better than 70% . FN is therefore partitioned by the *coverage* of the lot it belongs to, $\gamma = |R_i \cap M|/|R_i|$:

Both cut points are conventions, as the 5 m band is, and both were varied. The lower one is not load-bearing: from $\gamma \leq 0.05$ to $\gamma \leq 0.20$ the whole-lot-missed share moves

Class	Coverage	Treatment
whole lot missed	$\gamma \leq 0.10$	a detection failure
partly detected	$0.10 < \gamma \leq 0.70$	partial failure
fringe of detected lot	$\gamma > 0.70$	boundary imprecision

between 22.3% and 29.9% of FN, and the unresolved whole-lot remainder between 2.0% and 2.9% of labelled area, so what is concluded from it does not turn on where it sits. The upper one is more consequential — at 0.60, 0.70 and 0.80 the fringe class takes 51.0%, 44.4% and 31.9% of FN — so that share is quoted with its threshold attached.

Only the first is treated as a detection failure, and is further attributed to subtraction-stage removal, rooftop labelling, containment within OSM buildings, or left unresolved.

Throughout, attribution is positional and is worded as such. A false positive lying on OSM industrial land is reported as *located on* industrial land; it is not a claim that each such polygon was individually confirmed to be a storage yard.

3.6 Error typology II: stratified sampling

Automated attribution leaves a residual in each direction that no reference layer explains. These residuals are characterised by stratified random sampling with visual adjudication (Table 3.2). The design follows the three-part structure recommended for accuracy assessment — a sampling design, a response design specifying how each sampled unit is judged, and an analysis that scales the sample back to the population (Olofsson et al., 2014; Stehman and Foody, 2019).

Table 3.2. Sampling design. Large polygons are deliberately oversampled; the ratio estimator corrects for this.

Population	Stratum (m ²)	Polygons	Area (km ²)	Sampled
Unexplained FP	100–300	604	0.1055	20
	300–1,000	327	0.1656	25
	> 1,000	61	0.1172	25
Whole lots missed	100–300	34	0.0079	10
	300–1,000	51	0.0292	15
	> 1,000	17	0.0377	17
OSM-only parking	100–300	97	0.0192	5
	300–1,000	117	0.0625	10
	> 1,000	59	0.1793	15
Total				142

Estimates use a stratified ratio estimator (Cochran, 1977). For stratum h of known total area A_h , with sample s_h and polygon areas a_i , the estimated area of category c is

$$\hat{A}_c = \sum_h A_h \cdot \frac{\sum_{i \in s_h, i \in c} a_i}{\sum_{i \in s_h} a_i}$$

so that oversampling of large polygons does not distort the totals. Intervals on these estimates come from a stratified bootstrap: the inspected polygons are resampled with replacement within each stratum and the estimator recomputed 5,000 times, with each replicate’s deviation scaled by $\sqrt{1 - n_h/N_h}$ so that heavily sampled strata contribute proportionately less spread — the largest missed-lot stratum, inspected in full at 17 of 17, contributes none. The intervals cover sampling variance only: not adjudication error, and not the frame exclusion described next.

The sampling frame excludes polygons below 100 m². For unexplained FP this removes 0.0513 km², or 11.7% of that population (0.4396 km² in total, 0.3883 km² in frame). Percentages reported for this population are therefore shares of the frame, and the corrections derived from them (§4.6) implicitly assume the excluded slivers have the same composition as the sampled material. The assumption is conservative in the direction that matters: were the slivers composed like the sampled polygons, the upward correction to precision would be *larger* than the one reported. Slivers below 100 m² are in any case more plausibly boundary artefacts than real parking, so they are not extrapolated to.

Each sample was inspected as an image chip cut from the Digimap tiles and assigned one category from a fixed list, with an optional note. Adjudication was carried out against the Digimap imagery, since a model’s failure can only be assessed against the imagery it was given. Categories are organised by failure *mechanism* rather than surface appearance — what visual cue was absent — so that the typology maps onto plausible causes rather than onto descriptions.

3.7 Ablation design

The contribution of each building and road subtraction is measured by a 2×2 factorial design: the pre-subtraction prediction, buildings removed only, roads removed only, and both. Because set difference is commutative in the relevant sense,

$$(X \setminus A) \setminus B = X \setminus (A \cup B)$$

order does not affect this design; order matters only for the exclusive partition of §3.5. As a check that the reconstruction is faithful, applying both subtractions in one operation is compared against the pipeline’s own tile-by-tile output; the two agree on all three reported measures to the displayed precision, so the ablation can be trusted to isolate these two operations.

A second set of variants tests the reference layers as filters rather than as explanations. Sports pitches, industrial and commercial land, and road buffers widened by a further 6 m are each subtracted from the finished map, singly and together. This is a deliberate test of a tempting inference: a layer that explains where errors fall might be assumed to improve the map if used to remove them. Reporting recall alongside precision for these variants is what makes the test informative.

Rooftop parking is examined separately as a case where the pipeline and the model can be shown to disagree. The sixteen labelled rooftop lots are compared against both the pre-subtraction and post-subtraction outputs.

3.8 Consistency of the two imagery sources

Labelling was carried out over a satellite basemap, whereas the model operated on Digimap aerial tiles. Differences between the sources could therefore contribute to the measured error. One spatial diagnostic and two checks of reference–imagery disagreement assess that contribution.

(i) **Spatial alignment.** For the 1,478 labelled lots larger than 300 m² and covered by the model by at least 70%, let $\mathbf{d}_i$ be the vector from the centroid of the labelled polygon to the centroid of the predicted area within it. A common directional displacement makes the mean vector large relative to the average displacement magnitude, whereas varying boundary errors tend to cancel. The ratio is

$$\rho = \frac{\|\bar{\mathbf{d}}\|}{\frac{1}{n} \sum_i \|\mathbf{d}_i\|}, \quad \bar{\mathbf{d}} = \frac{1}{n} \sum_i \mathbf{d}_i$$

with $\rho \rightarrow 0$ where the common directional component is small relative to the scatter, and $\rho \rightarrow 1$ where displacement vectors share a common direction. Here $\|\bar{\mathbf{d}}\| = 0.208$ m (0.83 pixels at 0.25 m resolution), against a mean displacement magnitude of 1.247 m (4.99 px), giving $\rho = 0.167$. Non-zero mean components and a non-uniform direction distribution are statistically detectable ($p < 0.001$), but with 1,478 lots the sub-pixel magnitude is more informative than significance. This diagnostic shows no large common directional displacement among well-detected lots, so gross mis-

registration is unlikely to dominate the measured boundary error.

(ii) **An unresolved residual.** The whole-lot FN attribution in §3.5 leaves 0.0699 km² unresolved after separating areas associated with post-processing removal, rooftop labels and containment within building footprints. This is 2.1% of labelled area. If the entire residual were treated as reference–imagery disagreement, recall would rise from 0.854 to 0.873. This is a sensitivity scenario for the unresolved residual, not a strict upper bound on all disagreement between the sources.

(iii) **A sampled estimate.** The stratified sample in §3.6 covers whole labelled lots with no more than 10% coverage in both the post- and pre-subtraction outputs, excluding tagged rooftops and polygons below 100 m²; its frame contains 0.0748 km² of labelled lot area. Adjudication against the model’s input imagery estimates 0.0313 km², or 1.0% of labelled area, as not parking in that imagery. In the reference-side correction used here, removing it raises recall from 0.854 to 0.863 and leaves precision unchanged.

Separately, last-edit timestamps were retrieved for all 985 OSM parking features to test whether OSM disagreement simply reflects stale records. The 11 sampled features with no parking visible in the imagery had a median last-edit year of 2025, the most recent of any sampled category. This does not support stale OSM records as a general explanation; timestamps record any edit, including tag-only changes, rather than construction.

3.9 The calibration estimator and its spatial grain

Adjusting mapped area for commission and omission error using reference data is established practice (Olofsson et al., 2014). Here, where a map has precision p and recall r , predicted area $|M|$ relates to reference area $|R|$ by

$$|M| \cdot p = |TP| = |R| \cdot r \quad \Rightarrow \quad |R| = |M| \cdot \frac{p}{r}$$

Applied to the same cells used to measure p and r , this is an identity rather than an independent estimate. The factor simplifies to an area ratio:

$$\frac{p}{r} = \frac{|TP|/|M|}{|TP|/|R|} = \frac{|R|}{|M|}$$

Fitting it therefore requires only labelled and predicted total area over the calibration cells, not an object-level error analysis. Its usefulness must be tested on cells not used

to fit it.

Three hold-out schemes test whether a factor fitted on one part of the city predicts another, and at what grain (Table 3.3). Each fits p/r on cells $\mathcal{T}$ and applies it to held-out cells $\mathcal{H}$. Whole cells, rather than polygons, are withheld because short-range spatial dependence means that randomly splitting dependent units can understate predictive error (Roberts et al., 2017). Error is reported relative to labelled area in the held-out set:

$$e = \frac{(\sum_{c \in \mathcal{H}} |M_c|) (p/r)_{\mathcal{T}} - \sum_{c \in \mathcal{H}} |R_c|}{\sum_{c \in \mathcal{H}} |R_c|}$$

Table 3.3. Hold-out schemes for the calibration factor.

Scheme	Held out	Tests	What it tests
Random half split	50 cells	200	transfer to a comparable area of the same city
Leave one distance band out	1 band	5	transfer across the urban gradient
Leave one cell out	1 cell	99	the finest grain the estimator could be used at

Per-cell true-positive area is recovered as $|TP_c| = p_c \cdot |M_c|$ before p and r are micro-aggregated as in §3.4. Directly averaging cell precision would instead use the macro value (0.5136 rather than 0.5708) and produce a mismatched factor. The cell with no labelled parking has undefined relative error and is excluded from leave-one-out, leaving 99. Because small labelled areas destabilise relative error and inflate its mean, median absolute error in km² is also reported.

Chapter 4

Results

This chapter reports the measured transfer. Section 4.1 gives the headline accuracy and how it varies across the city, answering the first half of RQ1. Sections 4.2 to 4.4 decompose the error and test the building and road subtractions, answering RQ2. Section 4.5 returns to spatial variation and shows the apparent location effect to be confounded. Section 4.6 reports the sampled corrections and what the reference data itself is worth. Section 4.7 answers RQ3 within the reliability the preceding sections establish.

4.1 Overall accuracy and its spatial variation

Against 3.2597 km² of labelled surface parking, the model predicts 4.8785 km² — 1.50 times the labelled area.

Table 4.1. Accuracy of the post-processed output over the 100 km² study area.

Aggregation	Precision	Recall	IoU
Micro, all confidence levels	0.5708	0.8543	0.5202
Micro, confidence 2–3 only	0.5287	0.8658	0.4886
Macro (mean of 100 cells)	0.5136	0.8468	0.4697

The pattern is asymmetric. Recall of 0.854 means the model finds most labelled parking; precision of 0.571 means that a little under half of what it returns is not labelled parking. Decomposed by area, TP is 2.7848 km², FP 2.0937 km² and FN 0.4749 km², the last being 14.6% of labelled area.

Figure 4.1 shows that the asymmetry is not an artefact of aggregation: recall is generally high and varies less across the study area, while precision ranges from below 0.3 to above 0.8 between neighbouring cells. The macro figures fall below the micro figures on all three measures — precision 0.514 against 0.571 — indicating that cells contributing little parking area perform worse than the area-weighted total suggests.

Restricting the reference to the more confident labels does not improve precision; it lowers it, from 0.571 to 0.529. Since removing labels can only convert true positives

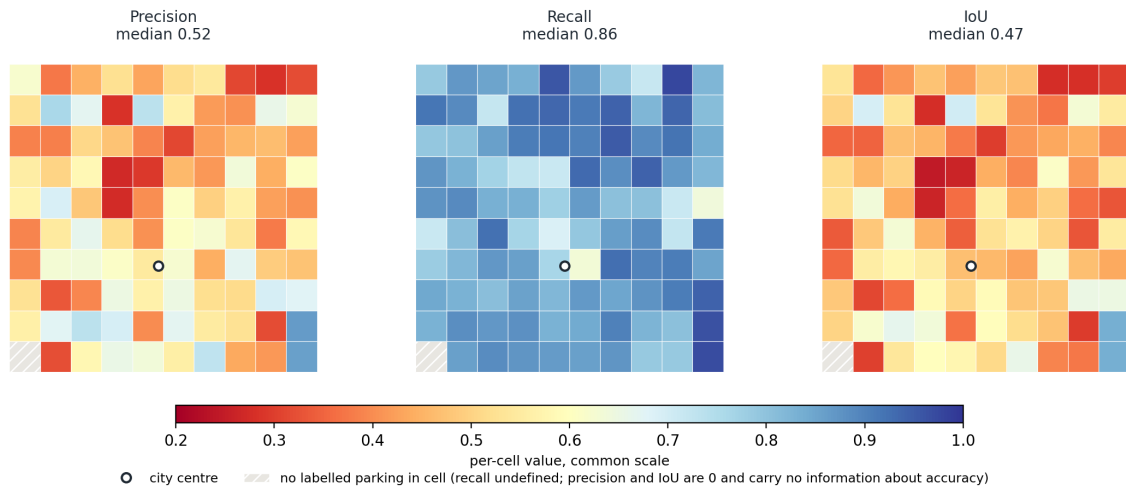


Figure 4.1. Precision, recall and IoU for each 1 km² cell, on a common colour scale. Recall is generally high and less spatially variable than precision. One cell contains no labelled parking and is hatched.

into false positives, this establishes that the confidence-1 labels are not a substantial source of the measured over-prediction.

4.2 False positives

False-positive area is 2.0937 km². Figure 4.2 shows what it is made of.

Boundary effects account for a substantial minority. False-positive area lying within a fixed distance of a labelled car park — the model finding the right lot and drawing it too large — is 17.3% at 2 m, 28.8% at 5 m and 36.3% at 10 m. The three thresholds are reported because 5 m is a working convention; the value chosen shifts this component by nearly twenty percentage points.

Attribution of the standalone remainder. Table 4.2 gives the exclusive partition alongside the unordered diagnostic.

Industrial and commercial land coincides with over half of all false-positive area. Buildings account for none, because OSM building footprints have already been subtracted at the post-processing stage; false positives on buildings the OSM data does not record fall into the unexplained residual. Moving industrial land from an early position in the peeling order to last changes its exclusive share from 30.0% to 29.6%, so the attribution is not an artefact of the ordering.

The unexplained residual divides into three unlike things. Stratified sampling of 70 chips (Figure 4.3) estimates that 44.5% of the residual is genuine misdetection — grey hardstanding, goods yards, sports courts, unpaved ground and unmapped houses — while 34.9% is real parking that the annotation rules deliberately exclude,

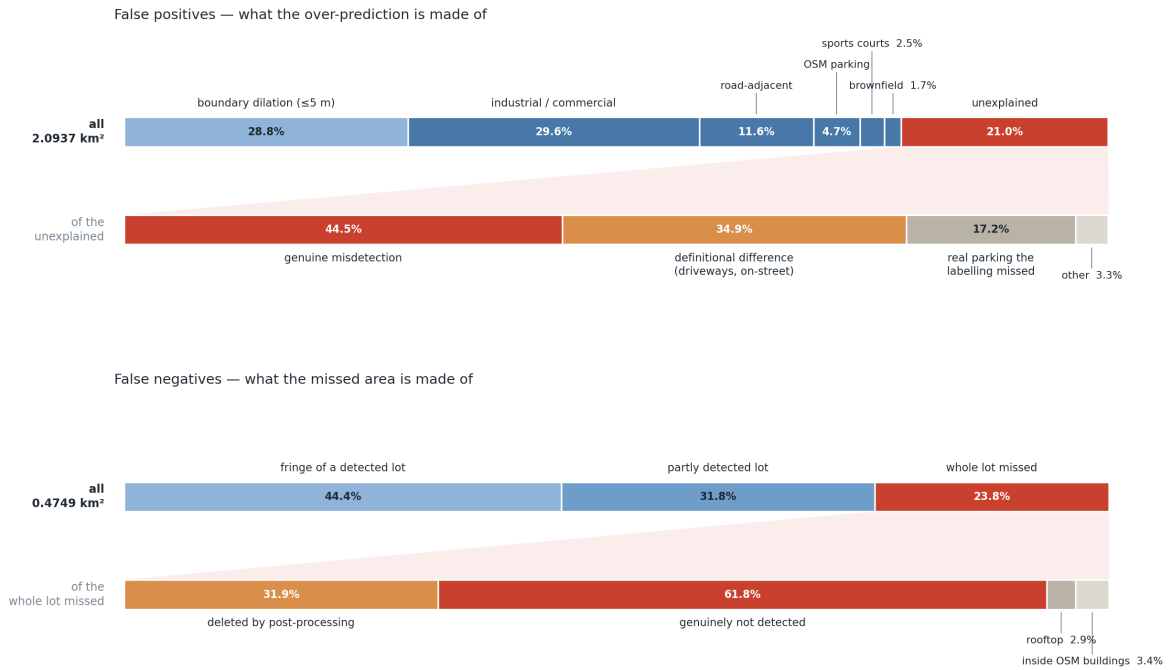


Figure 4.2. Composition of false-positive area (upper) and false-negative area (lower). In each case the lower bar expands the residual segment of the bar above it. Exclusive shares are of all FP or FN and sum to 100%.

Table 4.2. Where false-positive area falls. The two columns have different denominators and are not differences of one another: exclusive shares are assigned once each and sum to 100%, while unordered overlaps are measured against all FP independently and may overlap one another.

Layer	Exclusive (% of all FP)	Unordered overlap (% of all FP)
Boundary dilation (≤ 5 m)	28.8	—
Industrial / commercial land	29.6	52.8
Road-adjacent (+6 m)	11.6	16.9
OSM parking	4.7	9.1
Sports courts	2.5	3.1
Brownfield	1.7	2.3
Buildings	0.0	0.0
Unexplained	21.0	—

principally private driveways (20.2%) and on-street parking (14.7%). A further 17.2% is parking the labelling itself missed. Percentages are of the sampling frame defined in §3.6.

4.3 False negatives

Missed area is 0.4749 km², 14.6% of labelled parking. Measured against the prediction, 33.4% of it lies within 2 m of something the model did find, 54.1% within 5 m and 69.4% within 10 m.

Most missed area is the edge of a car park the model found. Classifying missed area by how much of its parent lot the model covered gives a more useful split than any distance threshold:

Table 4.3. Missed area by the state of the car park it belongs to.

Class	Coverage of the lot	Lots	Missed area (km ²)	Share of FN
Fringe of a well-detected lot	> 70%	1,638	0.2108	44.4%
Partly detected	10–70%	278	0.1510	31.8%
Whole lot missed	≤ 10%	121	0.1131	23.8%

Nearly half of all missed area belongs to lots the model detected to better than 70%. Only the third row is a detection failure in any useful sense, and it is not what it appears: 31.9% of it was found by the model and then deleted by post-processing, with a further 2.9% labelled as rooftop and 3.4% falling inside OSM building footprints. After these automated attributions, 0.0699 km² remains unresolved, or 2.1% of labelled area. The separate sampled assessment in §4.6 shows that reference–imagery disagreement also contributes to the whole-lot-miss population.

Detection tracks size and annotator confidence.

Table 4.4. Detection rate by lot size and by labelling confidence.

Lot size	Lots	Mean detection rate	Missed entirely
< 200 m ²	47	0.658	19.1%
200–500 m ²	563	0.763	8.2%
500–1,000 m ²	573	0.805	5.4%
1,000–2,500 m ²	559	0.852	3.0%
2,500–5,000 m ²	182	0.850	4.9%
> 5,000 m ²	113	0.864	1.8%

Lots below 200 m² are missed outright more than ten times as often as lots above 5,000 m². Detection is also lowest where the annotator was least certain, which indicates

Confidence	Lots	Mean detection rate	Missed entirely
1 (uncertain)	435	0.713	10.6%
2	1,137	0.829	3.8%
3 (clear)	465	0.856	5.4%

that part of the measured error reflects genuine ambiguity in the target rather than model deficiency.

What the unresolved whole-lot population looks like. Sampling of 42 chips estimates the composition of the residual as: not parking in the Digimap imagery 41.8%, irregular layout 23.3%, obscured by shadow or canopy 9.8%, unusual surface 9.2%, no cars present 6.3%, vans and lorries rather than cars 5.2%, and no markings 3.7%.

Irregular layout was identified in 11 of 42 chips and absent markings in one. Although the resulting 3.7% estimate for absent markings is imprecise, its interval remains well below that for irregular layout — 3.7% [0.6, 9.8] against 23.3% [16.9, 31.1]. The evidence therefore supports the ordering, rather than a precise ratio: irregular arrangement was the most frequently identified mechanism and unmarked surfacing among the least, revising the expectation in §2.5 that unmarked surfaces would be the principal difficulty.

4.4 Post-processing and the reference layers as filters

Table 4.5. Ablation. Variants A–D vary the building and road subtractions; E–H apply further layers as filters to the finished map.

Variant	Precision	Recall	IoU
A pre-subtraction output	0.5278	0.8939	0.4967
B – buildings	0.5467	0.8691	0.5051
C – roads	0.5498	0.8789	0.5111
D – buildings – roads	0.5708	0.8543	0.5202
E – sports pitches	0.5701	0.8373	0.5132
F – industrial land	0.4794	0.2789	0.2140
G – wider roads	0.5962	0.7884	0.5140
H – all three	0.5195	0.2419	0.1976

Reconstructing D from the pre-subtraction output in a single operation reproduces the pipeline’s own tile-by-tile result to within 0.0% on all three measures, so the ablation isolates the two subtractions.

The two subtractions together raise precision by 0.043 and lower recall by 0.040, for a net IoU gain of 0.024. Each contributes roughly half. Applied as filters, the further layers behave quite differently. Subtracting industrial and commercial land — the

Worked examples of each failure category, on the Digimap imagery the model saw
red = the sampled polygon · blue = the model's prediction · yellow = other labelled parking nearby



Figure 4.3. One worked example of each of twelve failure categories, on the Digimap imagery the model was given. Red outlines the sampled polygon, blue the model's prediction, yellow other labelled parking nearby.

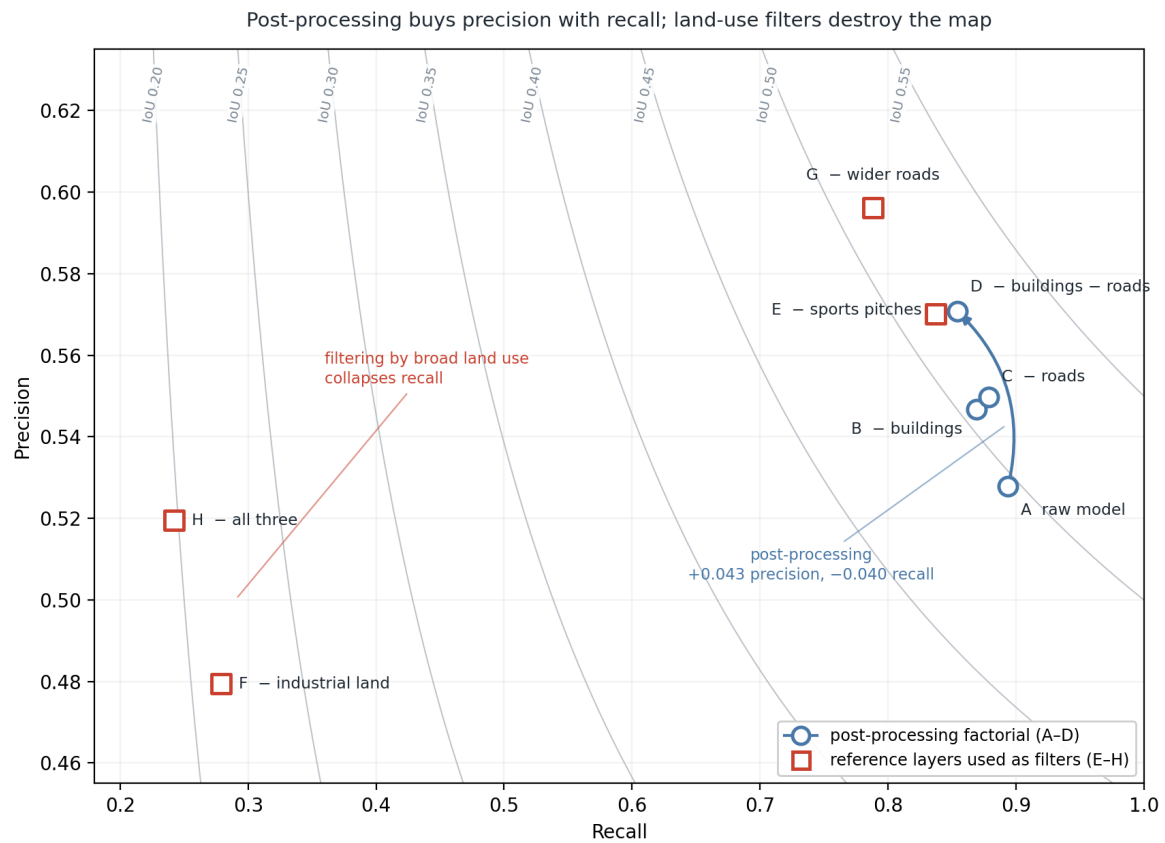


Figure 4.4. The eight variants on the precision–recall plane, with IoU iso-lines. Circles are the building-and-road subtraction factorial; squares are reference layers applied as filters.

layer that coincided with 52.8% of false-positive area — collapses recall from 0.854 to 0.279 and IoU from 0.520 to 0.214, because supermarket and retail-park car parks sit on exactly that land. Widening the road buffers is the only variant to raise precision (to 0.596), and it still lowers IoU.

The pipeline creates a blind spot of its own. Sixteen labelled lots, 0.0395 km² or 1.21% of the reference, are rooftop parking.

Table 4.6. Rooftop parking, before and after post-processing.

Measure	Value
Recall on rooftop lots, before subtraction	0.916
Recall on rooftop lots, after subtraction	0.115
Recall on non-rooftop lots, before subtraction	0.894
Rooftop area falling inside OSM buildings	85.6%
Rooftop area detected then removed	80.1%

The pre-subtraction output detects rooftop parking slightly *better* than ground-level parking. Subtracting building footprints removes four fifths of it.

4.5 Accuracy and location

Table 4.7. Correlations with per-cell precision, recall and IoU (Pearson r , p in brackets). Recall statistics use the 99 cells with labelled parking.

Metric	Distance	Parking share	Distance parking share	Parking share distance
Precision	−0.172 (0.087)	+0.536 (<0.0001)	+0.186 (0.065)	+0.540 (<0.0001)
Recall	+0.181 (0.073)	+0.103 (0.313)	+0.289 (0.004)	+0.250 (0.013)
IoU	−0.127 (0.208)	+0.515 (<0.0001)	+0.229 (0.022)	+0.540 (<0.0001)

Distance from the city centre does not predict precision. The share of a cell given over to parking does, strongly, and continues to do so after controlling for distance. The reverse does not hold: controlling for parking share, the distance effect is not significant. Distance and parking share are themselves correlated at −0.562.

All three metrics are reported to avoid selective presentation, but only precision is interpreted here. The partial correlations for recall and IoU reach significance while their raw correlations do not, and their signs reverse between the two; with $n = 100$ and two strongly correlated predictors this pattern is not a stable basis for a claim.

Read across distance bands, precision falls from 0.584 within 1 km to 0.485 beyond 4 km while recall stays between 0.70 and 0.86 (see Figure 4.5 and Appendix B). The band means move with parking share, not with distance.

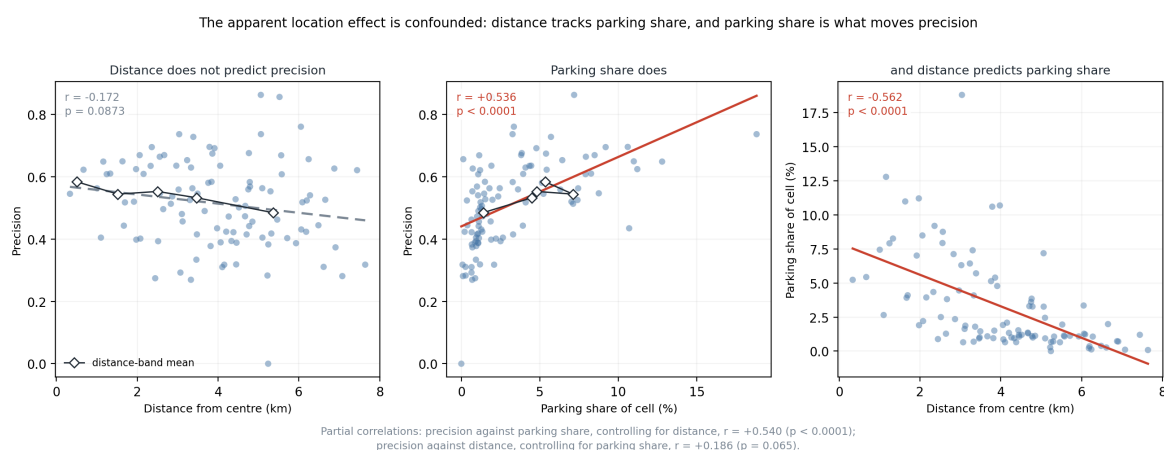


Figure 4.5. Per-cell precision against distance and against parking share, and parking share against distance. Solid red fits are significant at $p < 0.05$; the dashed grey fit is not. Diamonds are distance-band means.

4.6 Sampled corrections, and what the reference is worth

Applying the sampled estimates as corrections gives four cumulative variants:

Table 4.8. Accuracy under cumulative correction.

Variant	Reference (km ²)	Prediction (km ²)	Precision	Recall	IoU
1 As measured	3.2597	4.8785	0.5708	0.8543	0.5202
2 + reference-side (-0.0313 , not parking in the imagery)	3.2284	4.8785	0.5708	0.8626	0.5233
3 + prediction-side ($+0.0667$, parking the labelling missed)	3.2951	4.8785	0.5845	0.8654	0.5358
4 Effective (-0.1358, definitional exclusions removed)	3.2951	4.7427	0.6012	0.8654	0.5498

Across the four variants, precision rises from 0.571 to 0.601, about half of the gain coming from the final removal of on-street parking and private driveways — real parking excluded by the target definition rather than model error. Recall changes only from 0.854 to 0.865. For variant 4, the stratified bootstrap gives 95% intervals of $[0.5941, 0.6090]$ for precision, $[0.8635, 0.8675]$ for recall and $[0.5434, 0.5569]$ for IoU; its precision interval remains above the measured 0.5708. The headline pattern is therefore unchanged, and variant 1 remains the primary result.

OpenStreetMap, assessed against the same reference. OSM records 1.7641 km² of parking across 985 polygons, against 3.2597 km² across 2,037 labelled polygons. It overlaps only 1.1882 km², or 36.5% of labelled parking, leaving 63.5% absent from OSM. Similar median polygon areas (763 and 799 m²) suggest that this incompleteness is not explained simply by polygon size. Within the 0.2610 km² sampling frame of OSM parking polygons above 100 m² and no more than 10% covered by the labels, 63.2% is estimated to show no parking in the model-input imagery. The 11 sampled features in that category had a median last-edit year of 2025, so stale OSM records are not supported as a general explanation, subject to the timestamp limitation in §3.8.

4.7 The extent and distribution of surface parking

Table 4.9. Surface parking as a share of land, by distance band.

Band	Cells	Labelled	Model	Calibrated (km ²)
< 1 km	2	5.33%	6.42%	0.086
1–2 km	11	7.11%	10.35%	0.761
2–3 km	14	4.80%	7.19%	0.673
3–4 km	22	4.50%	6.53%	0.959
> 4 km	51	1.39%	2.29%	0.781
Whole area	100	3.26%	4.88%	3.2595

The calibrated column applies the whole-area factor from §3.9. Its whole-area value is an identity; the band rows show residuals when that citywide factor is applied within each band (−19.6% to +10.0%), not an independent transfer test.

The complete-coverage reference contains 3.2597 km² of surface parking, or 3.26% of the 100 km² study area. The sampled reference corrections in §4.6 raise this to 3.2951 km² (3.30%), a shift of less than 0.1 percentage points.

Parking share peaks at 7.11% in the 1–2 km band and then falls monotonically to 1.39% beyond 4 km. The apparent central dip to 5.33% is uncertain because the <1 km band contains only two cells. Across cells, the distribution is strongly right-skewed: mean 3.26%, median 1.71%, maximum 18.81%, with six cells above 10%.

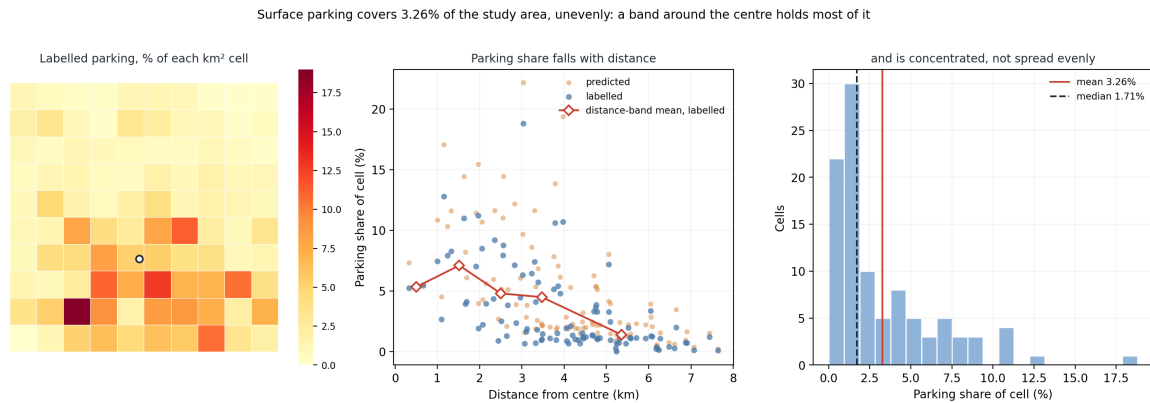


Figure 4.6. Labelled parking share by cell (white circle: City Square), labelled and predicted shares against distance, and the distribution of labelled shares across the 100 cells.

Calibration and the grain at which it holds. For a map with precision p and recall r , reference-equivalent area is estimated as predicted area $\times p/r$ (§3.9); here, $4.8785 \times p/r = 3.2595$ km². This is an identity on the cells used to fit the factor. Table 4.10 tests it on held-out cells.

Across the 200 random half splits, the central 90% of held-out errors runs from −6.6% to +7.8%, showing that a factor fitted on half the city transfers to a comparable sub-

Table 4.10. Error of the calibrated estimate on cells excluded from fitting the factor.

Scheme	Held out	Tests	Mean error	5th–95th pct.	Within $\pm 25\%$
Random half split	50 cells	200	+0.3%	–6.6% to +7.8%	100%
Leave one distance band out	1 band	5	–2.9%	–16.9% to +10.6%	—
Leave one cell out	1 cell	99	+16.5%	–22.0% to +80.5%	63%

area within the same city. This is not an uncertainty interval for the directly labelled 3.26%. At 1 km², the estimator becomes unstable: only 63% of cells fall within $\pm 25\%$, and the +16.5% mean is inflated by small denominators (median absolute error 0.0035 km² against mean labelled area 0.0329 km²).

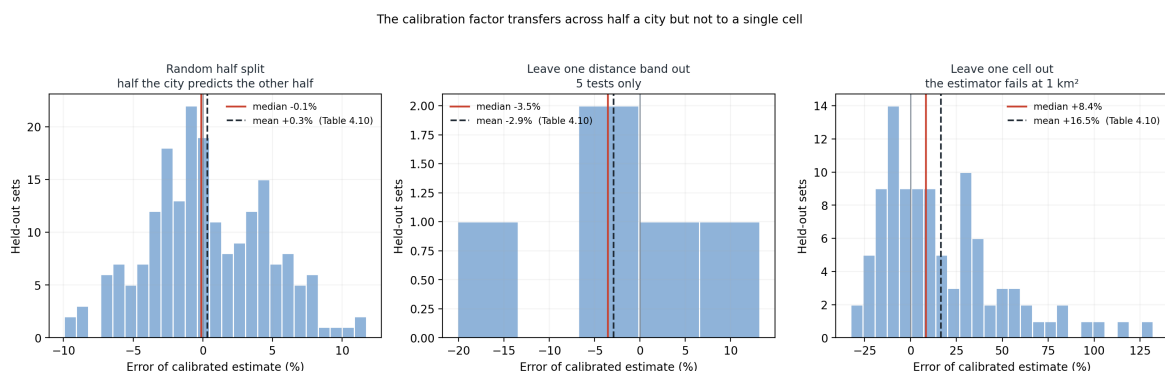


Figure 4.7. Held-out calibration errors under the three schemes. Red lines mark medians and dashed black lines means; the distance-band panel contains only five tests.

Chapter 5

Discussion

5.1 Answering the research questions

RQ1 — How accurate is the model on UK aerial imagery, and does accuracy vary systematically within the city?

The transfer works, but unevenly. Recall of 0.854 (Table 4.1) means that most labelled parking is found; precision of 0.571 means that a little under half of the returned area is not labelled parking, and predicted area is 1.50 times labelled area. Figure 4.1 shows the asymmetry across the city: recall is generally high and less spatially variable, while precision varies by more than a factor of two between neighbouring cells.

The IoU of 0.520 is better read against transfer studies than against an implicit standard of perfection. Maggiori et al. (2017) regarded roughly 65% IoU on unseen-city building footprints as satisfactory generalisation, while Hurst-Tarrab et al. (2020) report parking-block models above 50%. The present result lies in that range despite using no target-domain training in the primary analysis; the comparison studies drew training and test data from the same source. Scalar performance therefore survived the crossing better than might be expected, but it does not reveal where the cost fell.

Nor does the apparent spatial pattern survive closer examination. Once parking share is controlled for, distance from the centre no longer predicts precision, whereas parking share remains strongly associated with it (Table 4.7, Figure 4.5). Accuracy tracks parking abundance rather than location, consistent with macro scores falling below micro scores and sparse cells performing poorly. Lot size shows a separate association (Table 4.4); neither relationship establishes causation. The raw distance correlation alone would therefore give a plausible but misleading account of central-city performance.

RQ2 — What systematic errors does the model make, and how much error does post-processing remove or create?

Complete-lot non-detection is not the dominant explanation. Automated attribution

leaves 0.0699 km² unresolved, 2.1% of labelled area and under 3% of all error area; sampled adjudication also finds reference–imagery disagreement within this population. Boundary placement is the largest single component: 28.8% of false-positive area and 54.1% of false-negative area lie within 5 m of a matched feature, together about a third of all error area. Consistently, 44.4% of missed area belongs to lots detected to better than 70% (Tables 4.2, 4.3). The remainder mixes look-alike surfaces, definitional disagreement and pipeline artefacts rather than one dominant cause.

This qualifies the zero-shot robustness reported for SegFormer by Xie et al. (2021). Their benchmark degrades familiar scenes through noise, blur, weather and digital artefacts; geographic transfer changes the scene itself. Here, complete-lot detection appears comparatively robust, while boundary placement is the larger constraint. No comparable source-domain decomposition exists, so the study cannot say that delineation became worse after transfer. It can say that the architecture’s corruption robustness and the error pattern observed here are not in tension.

The sampled typology also revises the expectation in §2.5 that absent markings would be the principal difficulty. Irregular layout was identified in 11 of 42 chips and accounted for an estimated 23.3% of the unresolved whole-lot population, against one chip and 3.7% for absent markings. The latter estimate is imprecise, so the evidence supports an ordering rather than a precise ratio: irregular arrangement was the most frequently identified mechanism and unmarked surfacing among the least. Shadow, unusual surfacing and commercial vehicles appeared at more modest magnitudes.

Building and road subtraction is a genuine trade: it raises precision by 0.043 and costs 0.040 of recall (Table 4.5). It also creates a specific blind spot. The pre-subtraction output detects rooftop parking at 0.916 recall, but building subtraction removes four fifths of it; 31.9% of whole-lot misses were detected before being deleted by the pipeline (Table 4.6). That component is about half the size of the unresolved whole-lot remainder. Domain-adaptation research usually treats the model as the object transferred (Lyu et al., 2025), but deployment transfers a model and its corrections. The rooftop result therefore argues for re-evaluating both.

The ablation rejects another tempting inference. Although industrial and commercial land coincides with 52.8% of false-positive area, using it as a filter reduces recall from 0.854 to 0.279 and IoU from 0.520 to 0.214 because real retail parking occupies the same land. All four additional filters lower IoU, even where the loss is small. Reference layers can explain where error falls without being able to remove it.

RQ3 — How much central-city land is surface parking, and where is it concen-

trated?

The complete-coverage reference records surface parking over 3.26% of the 100 km² study area, rising to 3.30% after the sampled reference corrections (Table 4.8). Parking peaks at 7.11% in the 1–2 km band and declines to 1.39% beyond 4 km; the <1 km band contains only two cells, so the apparent central dip is uncertain. Distribution is strongly right-skewed: a median cell holds 1.71%, while six exceed 10%.

Jiao (2015) shows urban land density generally declining with distance from the centre. Parking does not show a clear decline within the inner 2 km, only beyond it. This is not a contradiction—the quantities differ—but it raises a question the data cannot answer: whether surface parking is displaced from the highest-value centre while remaining close to it. Testing that would require land-value data and more central cells.

5.2 What the transferred map is good for

The practical question is not whether 0.571 is a good precision but what a map of that precision supports—the fitness-for-use question raised in §1.1, for which quality depends on both the data and the task (Devillers et al., 2007).

It supports screening for candidate parking locations. Generally high recall means that most labelled parking is detected, but low precision means that returned locations require checking before site-level use.

It supports area estimation after local validation, at a stated grain. Across 200 random half splits, the central 90% of held-out errors runs from –6.6% to +7.8%; at 1 km², only 63% fall within $\pm 25\%$. The evidence therefore supports totals over comparable sub-areas within the city, not individual grid-cell values. Evidence between those scales is thinner: five held-out distance bands show errors from –20% to +13%.

For area calibration, p/r reduces to labelled area divided by predicted area. Testing the approach in a second city would therefore require labelled total area over sampled cells, not a full object-level error analysis—although transfer between cities remains untested. The map does not support uncorrected area measurement or site-level judgement: its raw area is 1.50 times the labelled total, and part of the apparent excess reflects the annotation definition rather than model error.

5.3 What the error decomposition adds

Three methodological observations follow, each from a single case.

First, equal IoU need not imply equal fitness for use. Cross-city benchmarks conventionally report per-city IoU (Maggiori et al., 2017), but that scalar conceals whether error comes from complete omission or imperfect boundaries. Here, complete-lot detection was comparatively robust and boundary placement contributed more error; the opposite profile at the same IoU would support different tasks. The distinction can be recovered from the spatial reference already required for segmentation validation.

Second, one accuracy figure can combine problems needing different remedies. Domain differences are candidates for adaptation (Lyu et al., 2025; Hong et al., 2023), whereas the 34.9% of unexplained over-prediction assigned to on-street parking and private driveways reflects the annotation scope. Reporting measured and effective precision separately prevents that definitional disagreement from being mistaken for model incapacity.

Third, area correction also has a spatial grain. Olofsson et al. (2014) show how reference data can adjust mapped area and quantify sampling uncertainty; the hold-out design here asks how far a locally fitted factor carries. It transferred across comparable sub-areas within Leeds but became unstable at 1 km². The result is case-specific, but the test is reusable.

5.4 Implications for UK parking evidence and densification

Bates and Leibling (2012) identify the absence of basic parking information as an obstacle to coherent British policy. Existing inventory routes do not transfer easily: American studies depend on cadastral records and codified parking requirements unavailable nationally in comparable UK form (Scharnhorst, 2018; Hoehne et al., 2019), while the British ground-survey approach has not been repeated at scale since the 1999 London exercise. Imagery offers a third route. It also measures area directly, unlike space counts that require assumptions about layout and aisle provision before they can answer a land-use question.

Where that area lies matters. The highest parking shares occur within 2 km of the centre, overlapping the inner band that Centre for Cities identifies as important to Britain's density gap (Lange, Kovacevic and Johnson, 2026). This is a spatial coincidence, not evidence that parking causes the deficit. The narrower conclusion is that the band where the density deficit is concentrated is not short of surface parking.

Concentration also qualifies Shoup's (2005) aggregate opportunity-cost argument. A citywide share of 3.26% does not reveal whether parking is diffuse or concentrated; here the median cell holds 1.71%, while six exceed 10%. For policy seeking under-

utilised land (MHCLG, 2024, para. 125(d)), identifying concentrations at supported aggregate scales is more useful than the headline share alone.

The OpenStreetMap comparison adds a smaller result to the VGI quality literature. OSM completeness is known to vary by place and feature type (Haklay, 2010; Sehra, Singh and Rai, 2013; Zhou, Wang and Liu, 2022); here, 63.5% of labelled parking is absent. Similar median polygon areas suggest that incompleteness is not explained simply by object size, but cannot establish the shape of the full size distribution. Separately, the sampled OSM polygons showing no parking in the model-input imagery were recently edited. Subject to the fact that timestamps record any edit rather than construction, this disagreement sample is not dominated by untouched legacy records.

5.5 Limitations and future work

One city. Hold-outs test transfer within Leeds, not between cities. Because the calibration factor is an area ratio, a multi-city test needs labelled totals over sampled cells rather than a repeat of the full object-level analysis, but that test has not yet been made.

A single annotator. Detection is lower for low-confidence than high-confidence labels (0.713 against 0.856, Table 4.4), indicating where visual or definitional ambiguity is concentrated. It cannot quantify annotation error or an accuracy ceiling without independent labels. Multiple annotators and an agreement measure are required (Foody, 2002).

Two imagery sources. Labels were drawn over a satellite basemap, while predictions used Digimap tiles. The unresolved whole-lot residual is 2.1% of labelled area and sampled adjudication gives a 1.0% reference-side correction (§3.8). The spatial diagnostic finds no large common displacement, but none of these checks eliminates source disagreement; labelling on the model-input imagery would.

Conventions shape the figures. The 5 m boundary band is not a natural break, so 2 m and 10 m are also reported. The sampling frame excludes fragments below 100 m² and covers 0.3883 km² of a 0.4396 km² residual. Reference-layer attribution is positional: a false positive on industrial land has not necessarily been verified as a storage yard.

The target definition matters. On-street parking and private driveways form 34.9% of unexplained over-prediction. They are real parking excluded by rule, so another defensible protocol would produce a different precision from the same map. Ap-

pendix A therefore reports the scope in full.

The typology suggests one training-data hypothesis: include more irregularly arranged car parks rather than simply more unmarked ones. With only eleven supporting chips, this is a direction to test, not a settled prescription.

Appendix C tests whether positional error categories can also guide training. Generic Leeds fine-tuning raises raw-pixel IoU from 0.485 to 0.614, trading recall for precision; targeted loss weighting adds no advantage that validation-selected thresholding of the generic model cannot match or exceed. The categories are therefore useful explanations but not, in this experiment, useful supervision. This single-split, one-seed result does not test visually targeted irregular layouts or transfer to another city. Its raw-pixel figures on 50 held-out cells are also not comparable with the post-processed results of Chapter 4.

Chapter 6

Conclusion

This dissertation tested whether a US-trained surface-parking segmentation model can be used to measure off-street surface parking in a British city. The model was applied to 100 km² of Leeds exactly as released, with no UK training data in the primary analysis, and evaluated against 2,037 manually labelled car parks drawn to the source model's own target definition. The error was then decomposed rather than merely reported: attributed exhaustively against independent reference layers, characterised by stratified sampling of 142 image chips adjudicated on the imagery the model actually consumed, and tested by ablation of the building- and road-subtraction stage.

The transfer works asymmetrically. Recall of 0.854 is generally high and less spatially variable than precision, which is 0.571; the model predicts 1.50 times the labelled parking area. Accuracy within the city tracks not distance from the centre but how much parking a cell contains — an apparent location effect that dissolves once parking abundance is controlled for.

Recognition failure is not the dominant explanation: automated attribution leaves 0.0699 km² of whole-lot FN unresolved, 2.1% of labelled area and under 3% of all error area, while sampled adjudication separately identifies reference–imagery disagreement. Boundary placement is the largest single component, at 28.8% of false-positive and 54.1% of false-negative area, with the remainder split between confusions with particular look-alike surfaces, disagreement over what counts as parking, and pipeline artefacts. Among the sampled unresolved whole-lot population, irregular arrangement was the most frequently identified mechanism and absent markings among the least. Building and road subtraction is a real trade, buying 0.043 of precision for 0.040 of recall, but it also creates a blind spot of its own: the pre-subtraction output detects rooftop parking better than ground-level parking, and building subtraction deletes four fifths of it.

Under that measured reliability, surface parking covers 3.26% of the study area as labelled, or 3.30% after the sampled corrections. The highest shares fall in the inner 2 km, reaching 7.11% in the 1–2 km band, with a clear decline beyond; and it is strongly concentrated: a median cell gives over 1.71%, while six exceed 10%.

Contributions. Four, in ascending order of transferability beyond this case. First, an empirical measurement of how a published US-trained parking segmentation model behaves in a British city, previously unavailable. Second, a reusable method for decomposing area-based segmentation error into boundary effects, attributable confusions, definitional disagreement and pipeline artefacts — a decomposition that turns a single accuracy figure into a statement about which uses survive. Third, evidence that a correction step justified by a sound premise can create a systematic blind spot, which is an argument for evaluating correction stages rather than assuming them. Fourth, a hold-out test of the spatial grain at which a standard area-calibration factor remains reliable: it holds to about $\pm 7\%$ at half-city scale within Leeds and fails at the scale of a single square kilometre.

Future work. The calibration factor reduces to a ratio of labelled to predicted area, so testing whether it transfers between cities requires labelled totals over a sample of cells rather than a repeat of the full error analysis undertaken here. This makes a multi-city test more tractable, although its cost and between-city validity remain untested. The typology points to a direction for training data: irregularly arranged car parks rather than simply unmarked ones, on the strength of eleven sampled chips. Imagery carrying a near-infrared band would test the one expected failure this study could not examine. A supplementary experiment (Appendix C) narrows what that validation should compare. Generic fine-tuning improved raw-pixel IoU while trading recall for precision; weighting the loss by positional error categories added neither overall nor selective gain, and its operating points were matched or bettered by adjusting the generic model’s threshold alone. The next question is whether, and which kind of, local supervision earns its labelling cost — a comparison of zero-shot use, area calibration, generic fine-tuning and visually targeted fine-tuning, judged on locating parking, estimating area, and the precision–recall trade-off each implies.

The Leeds results show that a transferred model cannot be trusted to measure parking area without local validation. Within the same city, local calibration supports area estimation at half-city scale; whether that finding transfers between cities remains open.

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Appendix A

Manual annotation protocol

These are the rules under which the 2,037 reference car parks of §3.2 were labelled. They are reproduced in full because the measured precision depends on where the scope line was drawn as much as on what the model can see (§5.5), so the protocol has to be inspectable rather than summarised.

A.1 Basis

The rules follow the annotation method of the US-trained model applied here (Qiam, Devunuri and Lehe, 2025). The same rules are used so that the accuracy figures are valid: ground-truth labels must follow the definition the model was trained on. Labelling was carried out in QGIS over the Google Satellite basemap.

A.2 Target

Off-street surface parking: open-air parking surfaces visible from above, outside public roads. Labels are binary (parking / non-parking). The use served is not recorded, as it cannot be judged reliably from imagery. No minimum-size threshold is applied — all off-street surface parking is labelled regardless of size, to match the definition the model was trained on.

A.3 Identifying parking

An area is labelled as parking where it has marked parking bays, or — where markings are absent — parked cars and a layout of bays and aisles that clearly show parking use. Areas whose use is unclear are left unlabelled or marked confidence 1.

Include

- Surface car parks, whatever use they serve.
- Parking bays and the internal aisles that connect them.
- Rooftop parking where the parking surface is visible from above.

Exclude

- On-street parking.
- Multi-storey or underground car parks with no visible parking surface.

Treated as non-parking, and also the common look-alikes the model most often confuses:

- Sports courts, storage or depot yards, and other non-parking hardstanding.
- Buildings, roads, pavements and landscaping.

A.4 Driveways

Only very short entrances belonging to a car park are included. Longer access roads are excluded, so that the labels do not teach the model to recognise roads (Qiam, Devunuri and Lehe, 2025).

Supplementary rule — residential parking. Individual single-house driveways or forecourts — a few cars, clearly one household — are not labelled: they are private curtilage, not a car park. Shared or communal residential parking courts serving several dwellings are labelled. This extends the source authors’ driveway rule rather than contradicting it: a single-house driveway is private access, like the driveways they exclude, whereas a communal court is genuine off-street surface parking under their target. This is the one point at which the protocol narrows the source definition, and the resulting difference is held separate in the error analysis of §4.2 rather than counted as model error.

A.5 Boundary

- Draw along the edge of the paving, not the parcel boundary.
- Keep one car park as one polygon, even where it is split by planting islands.
- Align to the current image. Where OpenStreetMap or another reference is out of date — a demolished building, for instance — follow the current image. OpenStreetMap parking is used only as a starting reference, never as ground truth.

A.6 Attributes

Field	Description
confidence	3 = clear, 2 = fairly clear, 1 = uncertain
notes	Optional note for ambiguous cases

The main validation uses the full label set. Results for the confidence 2–3 subset are reported alongside it in Appendix B as a sensitivity analysis, so that the effect of the uncertain labels is visible rather than assumed.

A.7 Why these rules, and how reliable they are

The model is validated against these labels, so the labels must match the definition the model was trained on. The rules therefore follow Qiam, Devunuri and Lehe (2025) as closely as possible: the same off-street surface target, pavement-edge boundaries, rooftop parking only where the deck is visible, and no minimum-size cut-off.

Annotating surface parking as a binary polygon class is an established approach in comparable aerial-imagery datasets — APKLOT (Hurst-Tarrab et al., 2020) and Grab-Pklot (Yin et al., 2022) are both built this way — so the approach is not ad hoc. APKLOT likewise fixes its target with an explicit include-and-exclude list, though it segments parking blocks rather than whole car parks, so the internal aisles labelled here fall outside its target; Grab-Pklot annotates whole car parks, the closer analogue to the target used here. The data were labelled by a single annotator, and no second labelling pass or inter-annotator agreement coefficient was produced; §5.5 treats this as a limitation, and Table 4.4 reports how far detection differs for the lots the annotator marked uncertain. Any point at which these rules differ from the source protocol is noted above, and error caused by such definitional difference is separated from model error in §4.2.

A.8 Sources

- Qiam, S., Devunuri, S. and Lehe, L.J. (2025) ‘A pipeline and NIR-enhanced dataset for parking lot segmentation’, WACV.
- Hurst-Tarrab, N., Chang, L., Gonzalez-Mendoza, M. and Hernandez-Gress, N. (2020) ‘Robust parking block segmentation from a surveillance camera perspective’, *Applied Sciences*, 10(15), 5364.
- Yin, Y., Hu, W., Tran, A., Kruppa, H., Zimmermann, R. and Ng, S.-K. (2022) ‘A context-enriched satellite imagery dataset and an approach for parking lot detection’, WACV, pp. 1371–1380.

Appendix B

Per-cell validation and accuracy by distance

Supporting tables for §4.1 and §4.5. The study area is a 10×10 grid of 1 km² cells; every cell is validated against the manual reference of Appendix A. all uses the complete label set, c23 the confidence 2–3 subset, so that the effect of the confidence filter is visible throughout.

B.1 Overall validation

Aggregation	Manual km ²	Model km ²	Model ÷ manual	Precision	Recall	IoU
Area-weighted, all labels	3.2597	4.8785	1.497	0.5708	0.8543	0.5202
Area-weighted, confidence 2–3	2.9790	4.8785	1.638	0.5287	0.8658	0.4886
Mean of cells, all labels	—	—	—	0.5136	0.8468	0.4697
Mean of cells, confidence 2–3	—	—	—	0.4756	0.8617	0.4408

B.2 Accuracy by distance band

Band (km)	Cells	Mean dist. km	Manual km ²	Model km ²	Mean parking share	Mean precision	Mean recall	Mean IoU
<1	2	0.503	0.107	0.128	0.053	0.584	0.698	0.462
1-2	11	1.52	0.782	1.139	0.071	0.545	0.828	0.489
2-3	14	2.504	0.672	1.007	0.048	0.553	0.86	0.509
3-4	22	3.471	0.989	1.436	0.045	0.533	0.837	0.485
>4	51	5.36	0.71	1.169	0.014	0.485	0.858	0.449

B.3 Correlations with distance and parking share

Partial correlations control for the other predictor. $n = 99$ for recall because one cell holds no reference parking and yields no recall value.

B.4 Per-cell results

All 100 cells, ordered by grid column then row. Cell identifiers follow the `c<col>r<row>` convention used in the project repository.

Test	n	Pearson r	p	Spearman ρ	p
distance vs prec_all	100	-0.172	0.0873	-0.186	0.0637
parking_share vs prec_all	100	0.536	0.0	0.653	0.0
distance vs prec_all controlling parking_share	100	0.186	0.0646	—	—
parking_share vs prec_all controlling distance	100	0.54	0.0	—	—
distance vs rec_all	99	0.181	0.073	0.161	0.1124
parking_share vs rec_all	99	0.103	0.3125	0.084	0.4097
distance vs rec_all controlling parking_share	99	0.289	0.0037	—	—
parking_share vs rec_all controlling distance	99	0.25	0.0127	—	—
distance vs iou_all	100	-0.127	0.2081	-0.16	0.1114
parking_share vs iou_all	100	0.515	0.0	0.639	0.0
distance vs iou_all controlling parking_share	100	0.229	0.0222	—	—
parking_share vs iou_all controlling distance	100	0.54	0.0	—	—
distance vs parking_share	100	-0.562	0.0	-0.66	0.0

Cell	Dist. km	Parking share	Model m ²	Manual m ² (all)	P	R	IoU	Manual m ² (c23)	P	R	IoU
c0r0	7.44	0.0120	15166.2	11958.1	0.6213	0.788	0.5323	11438.8	0.6	0.7956	0.5199
c0r1	6.65	0.0198	34517.6	19834.2	0.5264	0.9161	0.5022	14774.3	0.3933	0.9188	0.3801
c0r2	5.93	0.0108	22064.7	10779.8	0.3869	0.7919	0.3512	10475.2	0.3731	0.7859	0.3387
c0r3	5.30	0.0070	11121.2	7037.9	0.5534	0.8745	0.5127	6170.0	0.4827	0.8701	0.4502
c0r4	4.79	0.0100	15406.5	9984.8	0.5688	0.8777	0.527	9589.1	0.5546	0.891	0.5193
c0r5	4.46	0.0107	19572.6	10670.7	0.3885	0.7126	0.3359	10670.7	0.3885	0.7126	0.3359
c0r6	4.33	0.0095	18793.0	9459.0	0.394	0.7827	0.3551	9459.0	0.394	0.7827	0.3551
c0r7	4.43	0.0153	24464.7	15285.8	0.528	0.8451	0.4815	15285.8	0.528	0.8451	0.4815
c0r8	4.75	0.0361	53584.5	36125.2	0.5597	0.8302	0.5022	33897.9	0.5209	0.8234	0.4686
c0r9	5.24	0.0000	140.3	0.0	0.0		0.0	0.0	0.0		0.0
c1r0	6.91	0.0071	16400.9	7052.4	0.3736	0.8689	0.3537	7052.4	0.3736	0.8689	0.3537
c1r1	6.05	0.0335	39166.7	33469.1	0.7609	0.8904	0.6957	33105.2	0.7528	0.8907	0.6892
c1r2	5.24	0.0067	13932.1	6665.4	0.383	0.8005	0.3496	6665.4	0.383	0.8005	0.3496
c1r3	4.52	0.0120	19818.0	11971.5	0.5018	0.8307	0.4552	11971.5	0.5018	0.8307	0.4552
c1r4	3.91	0.0478	61374.4	47794.3	0.6919	0.8885	0.6366	47667.6	0.6916	0.8904	0.6373
c1r5	3.49	0.0145	21524.3	14536.1	0.546	0.8085	0.4835	13687.5	0.5079	0.7986	0.4502
c1r6	3.33	0.0409	53347.6	40889.4	0.6298	0.8217	0.5541	34852.4	0.5428	0.8309	0.4888
c1r7	3.46	0.0093	23073.3	9296.1	0.334	0.8289	0.3124	8714.3	0.3109	0.8231	0.2914
c1r8	3.86	0.0539	70603.4	53947.6	0.6745	0.8827	0.619	49100.9	0.6361	0.9146	0.6004
c1r9	4.45	0.0119	32059.6	11906.2	0.3188	0.8583	0.3028	9304.0	0.2773	0.9555	0.2738
c2r0	6.48	0.0038	7350.5	3841.8	0.4445	0.8505	0.4123	2803.6	0.3573	0.9367	0.3489
c2r1	5.56	0.0114	12243.9	11395.4	0.6691	0.7189	0.5304	11229.4	0.6613	0.721	0.5266
c2r2	4.67	0.0131	22189.7	13104.5	0.5067	0.8579	0.4675	10796.4	0.4089	0.8404	0.3794
c2r3	3.84	0.0150	19851.0	15031.1	0.584	0.7713	0.4978	12666.1	0.4883	0.7653	0.4247
c2r4	3.11	0.0163	28269.3	16284.5	0.4806	0.8342	0.4387	14123.0	0.4397	0.8802	0.4149
c2r5	2.56	0.0793	110231.0	79329.8	0.6645	0.9233	0.6297	71617.8	0.6047	0.9308	0.5787
c2r6	2.33	0.0434	59548.0	43401.3	0.6351	0.8714	0.5807	38202.4	0.5722	0.8919	0.535
c2r7	2.52	0.0250	51368.4	24978.6	0.3933	0.8088	0.3599	20950.9	0.3296	0.8082	0.3057
c2r8	3.04	0.1881	221784.2	188065.4	0.7369	0.869	0.6633	181524.9	0.7184	0.8778	0.6531
c2r9	3.76	0.0513	78303.1	51349.6	0.5808	0.8857	0.5403	41951.5	0.4872	0.9094	0.4647
c3r0	6.20	0.0036	5654.1	3561.5	0.5241	0.8321	0.474	3561.5	0.5241	0.8321	0.474
c3r1	5.22	0.0028	8997.8	2761.4	0.2832	0.9229	0.2767	2270.4	0.2444	0.9687	0.2425
c3r2	4.26	0.0100	19307.0	10044.1	0.4725	0.9083	0.451	7148.7	0.3574	0.9652	0.3529
c3r3	3.33	0.0069	18655.1	6909.9	0.2695	0.7277	0.2449	5584.4	0.2517	0.841	0.2403
c3r4	2.45	0.0088	26811.0	8825.9	0.2744	0.8336	0.2601	5764.0	0.1926	0.896	0.1884

Cell	Dist. km	Parking share	Model m ²	Manual m ² (all)	P	R	IoU	Manual m ² (c23)	P	R	IoU
c3r5	1.70	0.0410	60825.0	40990.3	0.5179	0.7685	0.448	37121.4	0.4885	0.8004	0.4354
c3r6	1.33	0.0826	115977.8	82639.9	0.6104	0.8566	0.5538	74318.9	0.5814	0.9072	0.5487
c3r7	1.64	0.1099	144298.8	109854.9	0.6497	0.8534	0.5844	96713.1	0.592	0.8833	0.549
c3r8	2.36	0.0919	116265.6	91920.3	0.6954	0.8795	0.6349	86667.6	0.6601	0.8855	0.6082
c3r9	3.24	0.0643	86001.4	64309.6	0.6535	0.874	0.5972	61277.8	0.6272	0.8803	0.5779
c4r0	6.06	0.0128	28404.9	12762.7	0.431	0.9593	0.4233	12586.9	0.4263	0.962	0.4192
c4r1	5.06	0.0326	41434.6	32632.8	0.737	0.9358	0.7015	32632.8	0.737	0.9358	0.7015
c4r2	4.06	0.0086	20280.0	8609.0	0.3896	0.9177	0.3764	8224.2	0.3818	0.9415	0.373
c4r3	3.07	0.0065	15801.8	6485.1	0.2926	0.713	0.2618	4163.0	0.1994	0.757	0.1874
c4r4	2.08	0.0221	42681.3	22054.8	0.4017	0.7775	0.3603	15420.9	0.3044	0.8424	0.288
c4r5	1.10	0.0265	45147.1	26483.0	0.4048	0.6902	0.3426	13962.9	0.1706	0.5516	0.1498
c4r6	0.34	0.0523	73194.4	52323.1	0.5455	0.7631	0.4665	49832.5	0.5233	0.7686	0.452
c4r7	1.01	0.0744	108407.4	74408.3	0.5631	0.8204	0.5013	69254.6	0.5235	0.8194	0.4693
c4r8	1.98	0.0190	39586.4	19005.5	0.3983	0.8295	0.3681	13944.5	0.3059	0.8685	0.2924
c4r9	2.97	0.0445	60551.9	44519.3	0.6357	0.8646	0.5781	40318.2	0.5859	0.8799	0.5425
c5r0	6.09	0.0122	20515.7	12247.5	0.5183	0.8682	0.4805	11091.4	0.4823	0.8922	0.4558
c5r1	5.09	0.0245	39582.4	24461.1	0.5634	0.9116	0.5342	23535.9	0.5451	0.9167	0.5193
c5r2	4.11	0.0065	18888.5	6508.2	0.3102	0.9004	0.2999	5635.7	0.2778	0.931	0.2722
c5r3	3.12	0.0187	38088.8	18675.5	0.4568	0.9316	0.442	17172.1	0.42	0.9316	0.4074
c5r4	2.16	0.0394	55963.4	39397.6	0.6093	0.8655	0.5566	32451.8	0.5248	0.905	0.4974
c5r5	1.25	0.0791	103149.1	79116.8	0.6085	0.7933	0.5252	77614.6	0.5982	0.795	0.5182
c5r6	0.67	0.0544	55216.3	54359.5	0.6228	0.6326	0.4574	49707.3	0.5591	0.621	0.4168
c5r7	1.16	0.1279	170632.8	127896.2	0.6489	0.8657	0.5896	122407.7	0.6195	0.8636	0.5643
c5r8	2.06	0.0849	106885.7	84901.7	0.6696	0.843	0.5954	76658.7	0.6051	0.8437	0.5442
c5r9	3.02	0.0631	96645.8	63078.9	0.5592	0.8568	0.5114	51522.8	0.4701	0.8817	0.4422
c6r0	6.28	0.0106	15457.0	10638.6	0.5407	0.7856	0.4712	9415.1	0.4786	0.7858	0.4234
c6r1	5.32	0.0107	24046.0	10676.7	0.418	0.9413	0.4073	9071.5	0.3667	0.9719	0.3628
c6r2	4.38	0.0066	14804.4	6562.2	0.4222	0.9524	0.4134	6250.5	0.4016	0.9512	0.3935
c6r3	3.48	0.0101	21438.4	10083.4	0.4152	0.8828	0.3935	9880.8	0.4073	0.8836	0.3865
c6r4	2.64	0.0128	20717.7	12773.6	0.4962	0.8048	0.4429	12422.4	0.4896	0.8166	0.4411
c6r5	1.97	0.1121	154368.5	112091.2	0.6247	0.8604	0.5672	110704.6	0.6174	0.861	0.5615
c6r6	1.67	0.0392	81924.5	39192.8	0.4432	0.9264	0.4281	36860.7	0.4195	0.9324	0.4072
c6r7	1.92	0.0701	114291.5	70136.6	0.5202	0.8476	0.4757	63122.6	0.4677	0.8469	0.4312
c6r8	2.57	0.0876	144495.0	87560.7	0.5471	0.9028	0.5167	82108.3	0.5193	0.914	0.4951
c6r9	3.39	0.0571	68407.9	57093.7	0.7282	0.8725	0.6582	46867.4	0.6224	0.9085	0.5857

Cell	Dist. km	Parking share	Model m ²	Manual m ² (all)	P	R	IoU	Manual m ² (c23)	P	R	IoU
c7r0	6.61	0.0026	6096.2	2645.5	0.3109	0.7165	0.2768	1963.5	0.2814	0.8737	0.2704
c7r1	5.71	0.0112	22925.0	11232.7	0.4032	0.8228	0.371	9185.3	0.3428	0.8556	0.3241
c7r2	4.85	0.0112	21847.6	11221.7	0.4561	0.888	0.4313	8724.3	0.3444	0.8624	0.3265
c7r3	4.05	0.0192	28391.2	19161.7	0.6356	0.9417	0.6115	17903.5	0.6039	0.9576	0.5882
c7r4	3.37	0.0179	25632.2	17915.9	0.5632	0.8057	0.4958	15607.6	0.5282	0.8675	0.4888
c7r5	2.87	0.0235	38779.4	23489.3	0.5371	0.8868	0.5027	22814.2	0.5267	0.8953	0.4962
c7r6	2.67	0.0381	51079.0	38122.6	0.6698	0.8975	0.6222	33867.7	0.6162	0.9294	0.5887
c7r7	2.83	0.0712	121748.1	71206.9	0.5135	0.8779	0.4792	66643.2	0.4805	0.8779	0.4504
c7r8	3.30	0.0740	116222.5	73997.6	0.5255	0.8254	0.473	66556.4	0.4704	0.8214	0.4268
c7r9	3.98	0.1069	193762.0	106906.7	0.4348	0.788	0.3892	99270.4	0.3987	0.7782	0.358
c8r0	7.08	0.0010	3554.0	1023.6	0.2813	0.9768	0.2795	1023.6	0.2813	0.9768	0.2795
c8r1	6.24	0.0012	1770.1	1237.9	0.6566	0.9389	0.6298	1237.9	0.6566	0.9389	0.6298
c8r2	5.47	0.0065	12892.8	6488.1	0.4627	0.9195	0.4447	6185.9	0.4511	0.9402	0.4385
c8r3	4.77	0.0110	21676.4	11033.1	0.4418	0.868	0.414	10725.7	0.4296	0.8682	0.4033
c8r4	4.20	0.0134	22488.5	13368.7	0.4239	0.7131	0.3622	12605.3	0.3989	0.7117	0.3434
c8r5	3.82	0.0095	18184.0	9507.3	0.3778	0.7226	0.33	8871.7	0.3626	0.7433	0.3223
c8r6	3.67	0.0109	20163.5	10875.2	0.4902	0.9088	0.4672	10597.4	0.4798	0.913	0.4589
c8r7	3.79	0.1060	138499.2	106006.9	0.696	0.9093	0.6508	92818.3	0.6171	0.9209	0.5861
c8r8	4.15	0.0209	52887.4	20879.2	0.3179	0.8053	0.2952	20879.2	0.3179	0.8053	0.2952
c8r9	4.71	0.0330	62589.3	32982.0	0.4151	0.7878	0.3734	22767.7	0.321	0.8824	0.3078
c9r0	7.64	0.0008	2038.2	785.8	0.318	0.8248	0.2979	785.8	0.318	0.8248	0.2979
c9r1	6.88	0.0073	9408.4	7265.2	0.6266	0.8115	0.547	7265.2	0.6266	0.8115	0.547
c9r2	6.18	0.0021	4157.5	2101.8	0.4236	0.838	0.3915	2101.8	0.4236	0.838	0.3915
c9r3	5.58	0.0108	14511.1	10764.1	0.6082	0.8199	0.5365	10588.4	0.5995	0.8216	0.5305
c9r4	5.10	0.0093	14750.9	9343.8	0.4049	0.6393	0.3296	8420.7	0.4024	0.7048	0.3443
c9r5	4.79	0.0327	51183.4	32712.9	0.5832	0.9125	0.5523	31635.8	0.5729	0.9269	0.5482
c9r6	4.67	0.0135	24329.6	13546.5	0.4733	0.8501	0.4368	13101.2	0.4626	0.8591	0.43
c9r7	4.76	0.0385	53547.4	38495.2	0.6762	0.9407	0.6486	38495.2	0.6762	0.9407	0.6486
c9r8	5.06	0.0718	80152.4	71847.7	0.863	0.9628	0.8352	71847.7	0.863	0.9628	0.8352
c9r9	5.52	0.0195	22122.4	19495.4	0.8564	0.9717	0.8356	19495.4	0.8564	0.9717	0.8356

B.5 Source files

Generated from `analysis/validation_summary.csv`, `analysis/accuracy_vs_distance.csv` and `analysis/accuracy_vs_distance_summary.csv` in the project repository (Appendix D).

Appendix C

Supplementary adaptation experiment

C.1 Purpose and standing

The primary analysis in Chapters 4 and 5 measures the released checkpoint applied to Leeds without any UK training data. This appendix reports a bounded supplementary experiment that departs from that boundary alone, in order to ask a question the main study cannot: if a user does hold local pixel-level labels, what does the error typology of §4.2 buy them?

Three interventions are compared against the zero-shot model: generic fine-tuning on Leeds labels, targeted fine-tuning that weights the loss by the attributed false-positive categories, and probability-threshold adjustment of the generic model with no retraining.

Nothing here enters the main analysis. All figures are raw pixel output on a fixed held-out split, without the post-processing that Chapter 4 applies throughout, and they are valid only for comparison with each other. They are not comparable with the headline figures of Chapter 4 and are not used to qualify them.

C.2 Data and split

The 100 cells of the study area are partitioned at cell level, so no patch from a training cell appears in evaluation:

Role	Cells	Use
Fit	40	gradient updates
Validation	10	epoch selection and threshold selection
Test	50	reported results only

Patch sampling keeps every patch containing labelled parking and an equal number of empty patches, giving 2,256 retained training patches. The validation set is the 438 retained patches of the ten validation cells — the same sample used to select checkpoints during training, so that threshold selection and epoch selection see identical data. The test set is the complete 3,200 patches of the fifty held-out cells, with

no retention filter, so evaluation covers whole cells rather than a parking-enriched sample.

The zero-shot arm reproduces precision 0.5190, recall 0.8819 and IoU 0.4853 on this split, which is the consistency check the other arms depend on: without it none of the comparisons below would be interpretable.

Tables C.1 to C.5 label the four arms A to D, following the result files listed in C.8. The text refers to them by description, so that arm letters are never confused with appendix letters.

C.3 Generic and targeted fine-tuning

Both fine-tuning arms start from the released checkpoint and share batch size, learning rate, optimiser and seed; only the loss differs. The targeted arm assigns per-pixel weights from the model’s own zero-shot errors on the fit cells, coded against the reference layers of §4.2: standalone false positives on precise layers (road buffer, curtilage, OSM parking, sports) are weighted most heavily, those on broad land-use layers less so, and false negatives receive a counterweight computed from the observed code composition rather than chosen by hand. False positives within the boundary band are deliberately left at unit weight, since upweighting them is precisely how a model is taught to draw everything smaller.

An earlier targeted configuration was run with a hand-set counterweight and is not reported here: its suppression and recovery terms were unbalanced, which confounds the comparison the arm was built to make. Only the rebalanced run is reported.

The two targeted checkpoints come from a single training trajectory, not from two independently trained models. Epoch 12 gave the best validation IoU (0.6105); among epochs within 0.02 of it, epoch 7 gave the highest validation recall (0.8385) at validation IoU 0.6093. The two differ by 0.0012 in validation IoU, which is within the epoch-to-epoch variation visible across epochs 7 to 12, so the pair should be read as two operating points on one trade-off curve rather than as a better and a worse model.

Table C.1. Overall accuracy, 50 held-out cells, raw pixels (micro)

Arm	Precision	Recall	IoU	Predicted ÷ reference
A zero-shot	0.5190	0.8819	0.4853	1.699
B generic fine-tuning	0.7664	0.7548	0.6136	0.985
C targeted, epoch 12 (best validation IoU)	0.7393	0.7168	0.5722	0.970
D targeted, epoch 7 (best validation recall)	0.6668	0.7852	0.5640	1.178

Table C.2. The same arms under macro (per-cell mean) aggregation

Arm	Precision	Recall	IoU
A zero-shot	0.4773	0.8777	0.4473
B generic fine-tuning	0.7553	0.7205	0.5833
C targeted, epoch 12	0.7116	0.6839	0.5374
D targeted, epoch 7	0.6391	0.7595	0.5322

Generic fine-tuning raises IoU from 0.485 to 0.614, gaining 0.247 of precision at a cost of 0.127 of recall: standalone false-positive area falls by 74.5% while false-negative area doubles. Neither targeted checkpoint improves on it, under either aggregation.

The generic model’s predicted area is within 1.5% of the reference total. This is arithmetic coincidence at this operating point, not a corrected bias: false-positive area (0.3736 km²) and false-negative area (0.3982 km²) happen to be close, and they cancel. The epoch-7 targeted checkpoint, from the same run, over-predicts by 17.8%. Nothing in this appendix supports a claim that fine-tuning corrects the area bias measured in Chapter 4, which concerns post-processed output over the whole study area.

Table C.3. Boundary decomposition at 5 m (km²)

Arm	FP dilation	FP standalone	FN erosion	FN standalone
A zero-shot	0.3737	0.9537	0.0858	0.1059
B generic fine-tuning	0.1302	0.2434	0.1684	0.2298
C targeted, epoch 12	0.1503	0.2602	0.1686	0.2913
D targeted, epoch 7	0.2359	0.4013	0.1254	0.2233

Under every arm the added false negative is majority standalone rather than erosion — whole car parks missed, not edges trimmed. Bands at 2 m and 10 m are reported in `boundary_bands_arms.csv` and do not change this ordering.

C.4 Selectivity

The targeted arm was designed to remove false positives from four categories specifically: road buffer, curtilage, OSM parking and sports. If positional weighting works as intended, the removal rate for those categories should exceed the removal rate for the others by more than it does under generic fine-tuning.

Neither targeted arm is more selective than generic fine-tuning. Per category, generic fine-tuning removes more standalone false positive than the epoch-12 targeted checkpoint on road buffer (81.4% against 78.0%), curtilage (75.8% against 70.6%) and OSM parking (42.7% against 40.1%); only sports favours the targeted arm (97.5% against

Table C.4. Standalone false-positive removal against zero-shot (%)

Arm	Targeted categories	Other categories	Gap (pts)
B generic fine-tuning	72.5	62.8	9.7
C targeted, epoch 12	71.6	63.2	8.4
D targeted, epoch 7	58.3	50.6	7.8

90.0%). Total standalone false-positive removal is also higher under generic fine-tuning (74.5% against 72.7% and 57.9%).

C.5 Threshold adjustment: method

The targeted checkpoints differ from the generic model mainly in where they sit on a precision–recall trade-off. That raises a question the fine-tuning arms cannot settle on their own: whether those operating points require targeted training at all, or whether they are reachable by moving the decision threshold of the generic model.

The released pipeline takes the argmax of the two output logits, equivalent to thresholding the parking probability at 0.50. The sweep replaces that rule and nothing else. No weights are retrained, and the generic checkpoint is used unmodified.

1. Sweep the parking-class probability threshold from 0.05 to 0.95 in steps of 0.01 (91 values) on the 10 validation cells.
2. Select thresholds on validation only, under four rules: the default 0.50; the best validation IoU; the threshold matching the epoch-12 checkpoint’s validation recall; and the threshold matching the epoch-7 checkpoint’s validation recall.
3. Lock each selected threshold, then evaluate once on the 50 test cells.

Test labels play no part in selection. At threshold 0.50 the sweep reproduces the generic model’s test figures exactly (0.7664 / 0.7548 / 0.6136), confirming that the sweep and the training evaluation share one pipeline.

C.6 Threshold adjustment: results

At the threshold matched to the epoch-12 checkpoint, the generic model gives 0.063 more precision and 0.033 more IoU at recall within 0.006. At the threshold matched to the epoch-7 checkpoint, it is higher on all three measures: precision by 0.056, recall by 0.010, IoU by 0.045. The ordering holds under macro aggregation, where the matched generic thresholds give IoU 0.5707 against the epoch-12 checkpoint’s 0.5374 and 0.5801 against the epoch-7 checkpoint’s 0.5322.

Threshold selection itself gains almost nothing: the best validation threshold, 0.48,

Table C.5. Threshold-adjusted generic model against the targeted checkpoints, test cells

Model and rule	Threshold	Precision	Recall	IoU
B generic, default	0.50	0.7664	0.7548	0.6136
B generic, best validation IoU	0.48	0.7590	0.7625	0.6139
B generic, matched to C’s validation recall	0.60	0.8020	0.7113	0.6051
C targeted, epoch 12	—	0.7393	0.7168	0.5722
B generic, matched to D’s validation recall	0.39	0.7227	0.7948	0.6090
D targeted, epoch 7	—	0.6668	0.7852	0.5640

improves test IoU by 0.0003 over the default. The released argmax rule is already close to optimal for this model, which is why the comparison is informative — the targeted checkpoints are being measured against a generic model that has not been tuned in its own favour.

The conclusion this supports is that positional targeted weighting produced no overall performance gain that threshold adjustment of the generic model could not supply. It does not support the stronger claim that the targeted model learned no new features: these are aggregate accuracy measures, and they cannot observe what the network represents internally.

C.7 Limitations

- **One seed, one weighting scheme.** Every arm was trained once. The epoch-to-epoch spread in validation IoU across epochs 7 to 12 is comparable to the gap between the two targeted checkpoints, so the experiment cannot separate a small real effect from run-to-run variation.
- **Positional proxies are not visual categories.** Standalone false positives were weighted by the layer they fall on. A false positive on industrial land may be a roof, a hardstanding, a road or a vehicle storage yard; these are one location but not one appearance, and weighting them together may present the network with no consistent feature to learn. This is the most likely explanation for the negative result and is untested.
- **Raw pixels only.** No arm passes through the post-processing of §3.3. Comparison with Chapter 4 is invalid in both directions.
- **Validation is a parking-enriched sample.** Epoch and threshold selection used the 438 retained patches of ten cells, not whole cells, so the selected operating points are tuned on a distribution denser in parking than the test cells.
- **One city, one annotator.** Fit, validation and test cells are all Leeds, labelled by the same annotator against the same protocol. Nothing here tests whether a fine-tuned model transfers to a second British city, which remains the precon-

dition identified in §5.5.

- **The threshold result is about aggregate accuracy.** It shows no measurable overall advantage from targeted weighting; it does not establish that the two models are equivalent, nor that targeted training is unproductive in general.

C.8 Files

Results are reproduced from `targeted-finetuning/Parking_targeted_run2/`: `evaluation_arms.csv` (Tables C.1–C.2), `boundary_bands_arms.csv` (C.3), `selectivity.csv` and `standalone_fp_by_category.csv` (C.4), `threshold_sweep/generic_threshold_selected.csv` (C.5), and `targeted_log.csv` for the epoch record. The notebooks are `run_targeted_colab.ipynb` and `threshold_sweep_colab.ipynb`.

Appendix D

Code and data availability

D.1 Repository

All code written for this study is archived at <https://github.com/hou1020/Parking>. The directories referenced elsewhere in this dissertation are:

Directory	Contents
calculate/	Agreement against the manual and OSM references, polygon filtering and result merging
analysis/	Validation, error attribution, sampling, ablation and calibration (§3.4–§3.9)
fine-tuning/*.gpkg	The manual reference labels and the 1 km ² grid
fine-tuning/	Generic fine-tuning of the released checkpoint (Appendix C)
targeted-finetuning/	Targeted loss weighting and the threshold sweep (Appendix C)
parking-lot-mapping-tool/	The released pipeline, with the UK-specific tiling, inference and post-processing written for this study (§3.3)

D.2 Imagery

Getmapping aerial photography supplied through Digimap: 100 tiles at 0.25 m ground sample distance, three visible bands, EPSG:27700. The tile identifiers and version suffixes needed to reorder the same coverage are recorded in `parking-lot-mapping-tool/output_files/tif_processing_progress.csv`, and every processing step from the raw tiles onward is reproducible from the code once the imagery is obtained under an equivalent licence.

Source	Use	Retrieved
OpenStreetMap building footprints, road centrelines	Post-processing inputs (§3.3)	25 June 2026
OpenStreetMap land use, brownfield, pitch, amenity=parking	Error attribution only (§4.2)	25 June 2026
Ordnance Survey Open Greenspace	Sports facilities in error attribution	—

D.3 Reference data

OpenStreetMap data are © OpenStreetMap contributors, available under the Open Database Licence. Ordnance Survey Open Greenspace is published under the Open Government Licence. Neither is used as ground truth; the distinction is set out in §3.1.

D.4 Reference labels

The 2,037 manually labelled car parks are held in the repository as GeoPackage, together with the 1 km² validation grid and the confidence attribute described in Appendix A. These are the labels against which every accuracy figure in Chapter 4 is measured, and they are original to this study.

D.5 Model

The segmentation network is the published checkpoint of Qiam, Devunuri and Lehe (2025), obtained from the authors’ release and used without modification in the primary analysis. The fine-tuned checkpoints produced for Appendix C are derived works of that release and are not redistributed; the training code and logs that generate them are in `fine-tuning/` for the generic arm and `targeted-finetuning/` for the targeted loss weighting and threshold sweep.

D.6 Result files

Each appendix table is generated from a file in the repository rather than transcribed:

Table	Source
Appendix B.1	<code>analysis/validation_summary.csv</code>
Appendix B.2–B.4	<code>analysis/accuracy_vs_distance.csv</code> , <code>analysis/accuracy_vs_distance_summary.csv</code>
Appendix C.1–C.2	<code>targeted-finetuning/Parking_targeted_run2/evaluation_arms.csv</code>

Table	Source
Appendix C.3	targeted-finetuning/Parking_targeted_run2/boundary_bands_arms.csv
Appendix C.4	targeted-finetuning/Parking_targeted_run2/selectivity.csv, standalone_fp_by_category.csv
Appendix C.5	targeted-finetuning/Parking_targeted_run2/threshold_sweep/generic_threshold_selected.csv

D.7 Reproduction

The city-wide inference and the fine-tuning experiments require a GPU and were run in Google Colab; the notebooks (run_finetuning_colab.ipynb, run_targeted_colab.ipynb, threshold_sweep_colab.ipynb) pin transformers==4.57.1, otherwise running against the Colab environment’s own package versions, and cache intermediate outputs, so a run interrupted partway resumes rather than restarting. All other analysis runs on CPU from the committed CSVs.

Appendix E

Research log

E.1 Note on this log

This log records the supervision meetings, tasks, problems and outputs of the dissertation project between April and August 2026. It draws on the notes of supervision meetings held with CASA and Centre for Cities (Section E.2) and on the commit history of the project repository, archived at <https://github.com/hou1020/Parking> (Section E.3).

Two points on the dates. Commit dates record when work was committed to the repository, which may lag when it was carried out; and the manual annotation described in Phase 4 was carried out in QGIS and stored outside the repository, so the dates for that phase come from the annotation files themselves.

E.2 Supervision meetings

Ten meetings were held between April and August 2026.

#	Date	Attendees	Discussed
1	27 Apr 2026	Esra, Clara (CASA); Rob, Maurice (Centre for Cities); Khalis, Jiayi	Kick-off. The project was defined as deep-learning detection of off-street surface parking in UK cities, motivated by identifying unallocated land with housing potential. Transferring a US-trained model to UK urban form was flagged as the main challenge. A bi-weekly online sync and a shared GitHub repository were agreed.

#	Date	Attendees	Discussed
2	5 May 2026	Esra, Clara	Progress update during the coursework period: aerial data had been downloaded from Digimap, but no model runs carried out. The 20 August deadline was confirmed and the CASA writing workshops announced.
3	19 May 2026	Esra, Clara	Model accuracy. Manual annotation and retraining were judged too compute-intensive at that stage, so spatial intersection with OpenStreetMap building polygons was recommended as post-processing. The urban-policy framing was restated over a computer-vision framing.
4	2 Jun 2026	Esra, Clara	First Leeds results on nine tiles: high recall but low precision, with supermarket roofs a characteristic false positive. Asked how many tiles should be validated manually, given about seven minutes of compute per image.
5	16 Jun 2026	Esra, Clara	Edge cases, in particular industrial and logistics yards being flagged as parking. Agreed to define explicit filtering criteria and a low-confidence class in the methodology, so that the output reflects sites with genuine redevelopment potential.
6	22 Jun 2026	Esra, Clara	Comparison with Ordnance Survey mapping, which also omits informal and unallocated parking. Asked to choose between a methodological error analysis and a land-potential and policy analysis for a single city, and to archive cleaned code on GitHub.
7	9 Jul 2026	Esra, Clara; Maurice (Centre for Cities)	Logistics check-in: the project would run mainly online, with a sync the following Friday.

#	Date	Attendees	Discussed
8	14 Jul 2026	Clara	The published, un-finetuned model transferred poorly to UK imagery, especially in distinguishing storage yards and lorry parks from ordinary car parks. Weighed retraining on my own annotations against further post-processing; compute and time would restrict any retraining to a very small area.
9	28 Jul 2026	Clara	Advised against pursuing further computer-vision accuracy. The dissertation was refocused on why a US-trained model transfers poorly to UK urban form, what the gaps in OpenStreetMap are, and what the detected areas imply for planning and housing policy.
10	11 Aug 2026	Esra	With the deadline approaching and writing not yet started, it was agreed that I would stop running models and first draft the research questions, motivation and literature review, and that the model's limitations across different national and urban contexts would be written up as a substantive analysis chapter rather than treated as a failure.

E.3 Project timeline

Phase 1 — Scoping and data acquisition (27 April – 14 May)

The topic was agreed at the kick-off meeting: applying a published deep-learning segmentation model to detect off-street surface parking in a UK city, with the policy aim of identifying land that is not allocated in local development plans but has housing potential. Aerial imagery was ordered and downloaded from Digimap. No code was written; taught-module coursework and an examination on 14 May took priority.

Phase 2 — Replicating the published pipeline and first UK transfer (15 – 29 May)

Nine commits.

The repository was initialised on 15 May with the published parking-lot mapping tool of Qiam et al. (2025), its trained checkpoint, and the core literature. The first obstacle was format rather than method: the tool expects georeferenced GeoTIFF input, whereas Digimap delivers JPEG images with separate world and metadata files, so `make_uk_geotiff.py` was written to convert them. The first inference was run on a single Edinburgh tile (`nt2774`).

On 29 May the study area moved to Leeds and nine tiles (`se2526`–`se2728`, a 3×3 block) were put through batch inference. Three defects surfaced and were fixed the same day: malformed output polygons, an incorrect coordinate reference, and a thread-pool worker limit that stalled batch runs.

Post-processing was added at the end of the same day: minimum-area filters at 500, 1,000 and 2,500 m², merging of per-tile outputs into a single layer, and removal of predictions falling on OpenStreetMap buildings and roads. Agreement against OSM parking polygons at each threshold gave the first quantitative sense of how the model performed on UK imagery.

Phase 3 — Validation infrastructure (2 – 25 June)

Six commits.

The 2 June meeting established the central problem: recall was high but precision was low, with the flat roofs of large supermarkets the characteristic false positive. The review of 16 June added a second systematic confusion — industrial and logistics yards, where vehicles cluster but redevelopment for housing is not realistic — and made clear that agreement with OSM was not a sufficient basis for evaluation, since OSM parking is itself incomplete.

Work therefore turned to manual ground truth. A validation boundary was defined on 16 June, and on 23 June `calculate_manual_agreement.py` and a first manually annotated validation layer were committed, producing metrics against manual labels rather than against OSM.

Two pieces of infrastructure were settled on 25 June: OpenStreetMap retrieval was switched to the Overpass API in `post_processing_uk.py`, and Colab outputs were redirected to Drive so that results survived session resets.

Phase 4 — Manual annotation and redirection (26 June – 2 August)

No commits. Dates from annotation files.

This is the longest stretch of the project without a commit, and the work in it was annotation rather than code. The annotation frame and sampling grid were built on 26 June (`leeds_boudry.gpkg`, `leeds_grid.gpkg`); the proposal was revised on 30 June; the validation annotation set was completed on 19 July (`leeds_manual_validation.gpkg`); the annotation rules were written down on 21 July (`Rules.md`); and the main annotation set was completed on 28 July (`leeds_manual.gpkg`), together with `validate_removal_vs_manual.py`, which tests the OSM-removal step against the manual labels.

Writing the rules down was the more consequential of these steps. Because the model is validated against these labels, the labels have to match the definition the model was trained on, so the rules were aligned to Qiam et al. (2025) — the same off-street surface target, pavement-edge boundaries, rooftop parking included only where the surface is visible, and no minimum-size threshold — with any deviation recorded so that errors caused by definition differences could be separated from true model errors.

Two meetings in this window shaped what followed: the report of poor transfer on 14 July, where retraining on my own annotations was weighed against further post-processing, and the redirection of 28 July away from accuracy and towards why the model fails in the UK context.

Phase 5 — City-wide run and error analysis (3 – 10 August)

Four commits.

With the redirection settled, inference was extended from the nine-tile test block to the full Leeds study area on 3 August — 100 tiles, 413 output files — and the emphasis shifted from improving predictions to characterising their errors. False-positive analysis was added the same day, using OS Open Greenspace among the reference layers, alongside the results of validating the removal step against manual labels.

On 5 August three analyses were added: false negatives broken down by class, an ablation of the post-processing steps, and accuracy as a function of distance from the city centre, the last addressing whether the model fails differently in the centre and the suburbs. A stratified sampling design for error inspection followed on 7 August, and on 10 August 142 image chips were cut for visual inspection, together with a full

mosaic and a parking-extent map.

Phase 6 — Calibration, fine-tuning and the start of writing (11 – 12 August)

Seven commits.

The morning of 11 August was spent on a calibration-transfer analysis, a co-registration check, and the first methods figures. A Colab fine-tuning notebook was then built and iterated through six commits in a single afternoon, using the manual annotations and grid produced in Phase 4.

The meeting later that day found that no writing had begun, and it was agreed to stop modelling and draft the research questions, motivation and literature review first. `02_background.md`, the first dissertation chapter in the repository, was committed the following day, 12 August, along with the results figures.

Phase 7 — Write-up and targeted fine-tuning (13 – 20 August)

Ten commits to date.

The remaining chapters — abstract, introduction, results, discussion and conclusion — and the reference list were committed on 16 August. A targeted fine-tuning experiment was set up and iterated the same day, with the calibration-transfer error table added and the methods and results figures regenerated. The remaining time before submission is given to the front matter, reducing the manuscript to the word limit, and proofreading.